

A machine learning model for spatiotemporal forecasting of infectious disease trends in Mexico

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Abstract. Infectious diseases remain a significant public health challenge, contributing substantially to the disease burden and emerging as a primary focal point in public health. The complex, unpredictable nature of novel viruses such as influenza A(H1N1) and SARS-CoV-2 has required substantial efforts to accurately predict infectious diseases. While vaccination is a primary defense against infectious diseases, challenges arise from limited vaccination resources, viral mutations, and antibiotic-resistant bacteria. Studying infectious disease dynamics has played a crucial role among researchers. In recent years, Machine Learning and Deep Learning techniques have gained popularity due to their ability to discover connections among large amounts of data. Several studies have addressed the correlations between the prevalence of infectious diseases and a wide number of factors, such as ecological and socioeconomic drivers. Other studies have proposed forecasting methods to detect and raise alarms about new outbreaks, focusing on either historic spatial or temporal data. Few proposals use spatial and temporal analysis to address the dynamics of infectious diseases. The current research proposal focuses on a machine learning model that selects and merges mortality data and influencing factors such as meteorological data, to use them as part of a forecasting model to provide an accurate prediction of infectious disease outbreaks in Mexico. Thus, enabling decision-makers to effectively perform resource allocation.

Keywords: Spatiotemporal analysis, infectious disease, machine learning, artificial intelligence, forecasting

1 Introduction

Infectious diseases (ID) have historically imposed a significant burden on human health. However, scientific and technological advances have led to effective preventive and therapeutic measures which have resulted in a reduction in the contribution of infections as a cause of morbidity and mortality. Nevertheless, the appearance of emerging pathogens, such as influenza A (H1N1)pdm09 and COVID-19 (SARS-CoV-2), have highlighted that infectious diseases continue to be a significant threat to public health. As such, lower respiratory tract infections are a leading cause of morbidity and mortality globally, ranked as the fourth leading cause of death, resulting in nearly 2.6 million deaths in 2019 (WHO, 2020). Influenza and respiratory syncytial virus (RSV) are two of the main pathogens causing Acute Respiratory Infections (ARI) (Pavia, 2011; Wong-Chew et al., 2017). According to the World Health Organization (WHO), seasonal influenza results in about 3 to 5 million cases of severe illness, and 650 000 deaths globally each year (WHO, 2023). Moreover, since 2020, the COVID-19 pandemic has resulted in an increase in mortality associated directly or indirectly with ARI. Despite control of infectious disease outbreaks remains a key challenge of public health, difficulties with developing and disseminating immunizations, viral mutations, and the presence of antibiotic-resistant bacteria undermine the efforts made (Jacobs, 2018).

Predicting IDs remains a key challenge, due to insufficient disease surveillance efforts, the absence of robust disease reporting systems, and delayed and incomplete sharing of epidemiological data (Charles-Smith et al., 2015). Machine Learning (ML) and Deep Learning (DL) techniques hold a significant edge over traditional approaches because they do not place substantial restrictions on the data. Unlike traditional methods, they do not require prior assumptions about data distribution and structure, a fixed number of input feature variables, or a comprehensive understanding of how input features influence the presence of disease (Chae et al., 2018).

Previous studies have addressed the relationship between ARI and a wide number of factors, such as meteorological and ecological factors. X. Wang et al. (2017) proposed a predictive model based on weather factors and influenza-like illness (ILI) data. By using statistical methods, such as Moran’s I, the hotspot analysis model, concentration analysis, and correlation analysis, Song et al. (2021) studied the characteristics of the spatiotemporal distribution of and factors influencing the spread of influenza. D. Wang et al. (2022) performed a systematic review on the impact of geoenvironmental factors on global COVID-19 transmission, including climate, human mobility, demographic factors, and human interventions. Ambient air pollution and

its relationship with incidence, prevalence, severity, and mortality of ARI have been addressed by several studies (Katoto et al., 2021; Kawano et al., 2022; Keshtkar et al., 2022; Sahoo, 2021).

In addition, several models have been proposed as key elements for potential early warning systems (EWS). Z.-x. Wang et al. (2022) proposed a Combinatorial Judgment Classifier (CJC) model to classify epidemic trends using machine learning methods, proposing an early warning for influenza outbreaks. Dong et al. (2021) developed a data-driven model, the minimum-spanning-tree-based dynamical network marker (MST-DNM) to monitor the process of COVID-19 transmission in Italy. By using a dynamic Bayesian network and vector autoregressive moving average model, Qiu et al. (2021) proposed a concept called spatiotemporal route to display the transmission directions of influenza and measure them. Kianfar et al. (2022) used a multilayer perceptron artificial neural network to examine the spatiotemporal influence of 75 explanatory variables on COVID-19 prevalence and mortality. González-Bandala et al. (2020) proposed a methodology composed of a data acquisition process and a computational model, providing an accurate prediction for ARI data one week ahead. Although the methodology proposed covers the data acquisition process and uses it to make accurate predictions, it still needs to be expanded to work for smaller territorial units (i.e. state or municipality) allowing a decomposition into three dimensions: endemic, epidemic within the city, and epidemic among cities. Few previous studies have applied spatial and temporal analysis in their predictions or simultaneously incorporated spatial and temporal variations in the predictive models (O’Shea, 2017).

The current work proposes a machine learning model designed to work with small territorial and large units, capable of selecting and merging potential influential factors and mortality data due to infectious diseases. Thus, allowing the prediction of infectious disease spatiotemporal trends in Mexico.

1.1 Workgroup

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1.2 Hypothesis

Infectious disease spatiotemporal trends can be accurately predicted using machine learning models based on meteorological and mortality data.

1.3 Research goal

To develop a machine learning model that utilizes meteorological and mortality data to forecast spatiotemporal trends of infectious diseases in Mexico

1.3.1 Specific objectives

1. To identify the available data sources relevant to the proposal.
2. To collect and prepare the identified data for modeling.
3. To perform a Literature Review addressing infectious diseases through machine learning approaches.
4. To apply computational models to forecast infectious disease trends.
5. To integrate the models and deploy them into a data visualization module.
6. To validate the final product against available state-of-the-art techniques.

2 Literature review

2.1 Planning the literature review

A Literature Review allows the identification, evaluation, interpretation, and synthesis of the available evidence relevant to a topic or research question. During the planning phase, the need to perform the review is identified, the research questions are defined, a set of search strings are created, and the data sources for primary study selection are selected.

2.1.1 Identifying the need to perform the literature review

Numerous studies explore the influential factors of infectious diseases and techniques for predicting trends and detecting outbreaks. While some focus solely on identifying influential factors, others utilize statistical methods or harness Machine Learning approaches for infectious disease forecasting. A thorough review of the current literature on infectious disease forecasting approaches is essential to grasp the state of the art, minimize the bias, and enable the proposal of either a proper enhancement to existing techniques or a new technique for ID forecasting.

2.1.2 Research questions

To gather most of the studies and to accurately compare the results from the execution of the literature review, The framework of Population, Intervention, Comparison, Outcome, and Context (PICOC) is applied to define the elements of the research question (Table 1):

1. How do existing computational models perform in terms of accuracy, precision, sensitivity, and specificity when predicting the number of cases and deaths due to infectious diseases in humans?

Table 1: Research question formulation through the PICOC framework

Population	Infectious Diseases
Intervention	Computational Models
Comparison	Accuracy, precision, sensitivity, and specificity
Outcome	Prediction of the number of cases and deaths
Context	Human population

2.1.3 Search strings

To streamline the search string creation process, the search concepts are derived from the research question, and corresponding synonyms for each search concept are identified, including the additional concept “F1-score”, as illustrated in Table 2.

Table 2: Search concepts and synonyms

Search concept	Synonyms
Case	Trend, Transmission
Infectious Disease	Infection
Computational Model	Machine Learning, Artificial Intelligence, Deep Learning, Artificial Neural Network
Prediction	Forecasting
Accuracy	Effectiveness, Performance
Sensitivity	Recall

Three databases are selected for the literature review: 1) Springer, 2) PubMed, and 3) Elsevier. The search concepts are combined with Boolean Operators “AND” and “OR”, and adapted to the corresponding databases, resulting in the search string presented in Table 3.

Table 3: Resulting search string.

Identifier	Search string
Springer, PubMed	(“Infectious disease” OR “Infection”) AND (“Cases” OR “Deaths” OR “Trend” OR “Transmission”) AND (“Human” OR “People” OR “Population”) AND (“Machine learning” OR “Artificial Intelligence” OR “Deep Learning” OR “Neural Network”) AND (“Prediction” OR “Forecasting”) AND (“Effectiveness” OR “Performance” OR “Accuracy” OR “Sensitivity” OR “Specificity” OR “Precision” OR “Recall” OR “F1-score”)
Elsevier	(“Infectious disease” OR “Infection”) AND (“Machine learning” OR “Artificial Intelligence” OR “Deep Learning” OR “Neural Network”) AND (“Prediction” OR “Forecasting”)

2.2 Conducting the review

PRISMA (Preferred Reporting Items for Systematic reviews and Meta-Analyses) guidelines were followed while conducting this review. In addition to the three datasets already selected, sixteen studies coming from a previous literature review are included (Table 4). The flowchart presented in Figure 1 is employed at different stages to ensure a thorough selection of research articles.

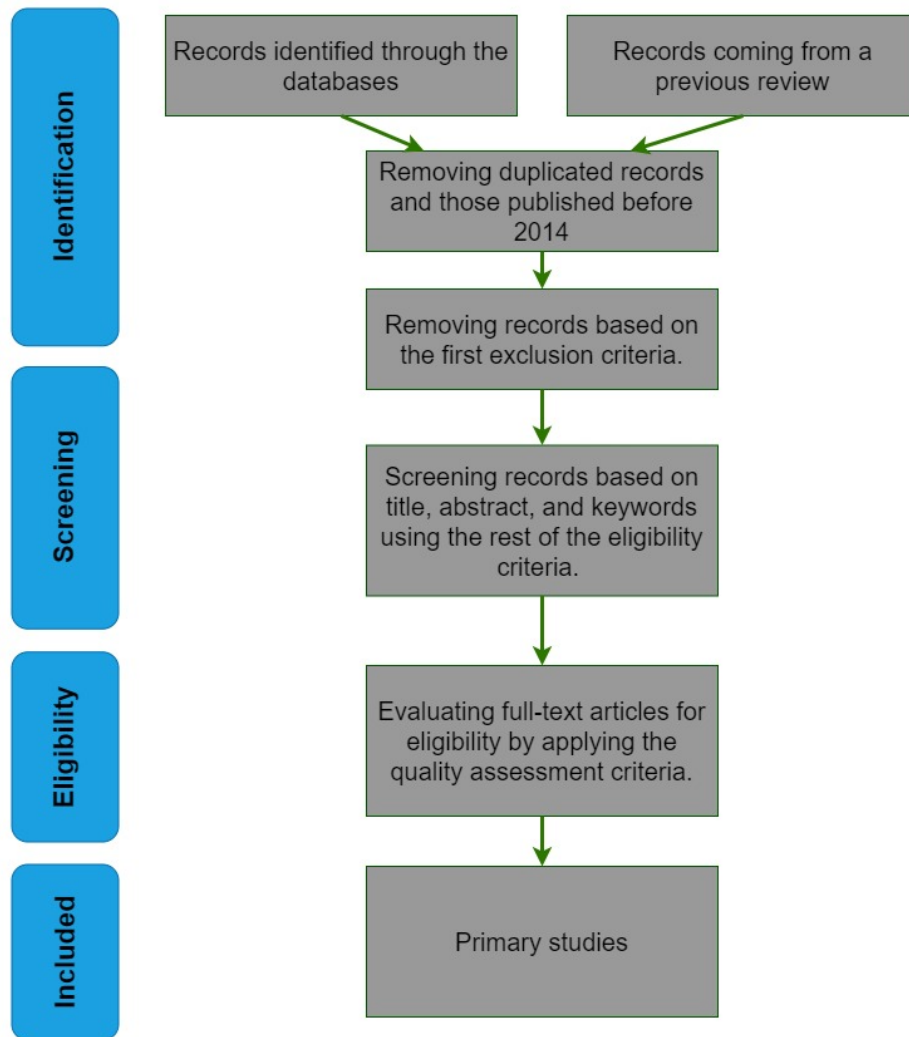


Figure 1: Flowchart of the literature search process

Table 4: Primary studies coming from a previous review.

Identifier	Title
1	The impact of geo-environmental factors on global COVID-19 transmission: A review of evidence and methodology
2	Spatio-temporal modeling of COVID-19 prevalence and mortality using artificial neural network algorithms
3	Peculiar weather patterns effects on air pollution and COVID-19 spread in Tokyo metropolis
4	Acute and chronic exposure to air pollution in relation with incidence, prevalence, severity, and mortality of COVID-19: a rapid systematic review
5	Research on epidemic spread model based on cold chain input
6	Construction of Influenza Early Warning Model Based on Combinatorial Judgment Classifier: A Case Study of Seasonal Influenza in Hong Kong
7	Spatio-temporal Differentiation in the Incidence of Influenza and Its Relationship with Air Pollution in China from 2004 to 2017
8	Significance between air pollutants, meteorological factors, and COVID-19 infections: probable evidences in India
9	How do temperature, humidity, and air saturation state affect the COVID-19 transmission risk?
10	Analysis of changes in air pollution quality and impact of COVID-19 on environmental health in Iran: application of interpolation models and spatial autocorrelation
11	GIS-based approaches on the accessibility of referral hospital using network analysis and the spatial distribution model of the spreading case of COVID-19 in Jakarta, Indonesia
12	Spatial transmission network construction of influenza-like illness using dynamic Bayesian network and vector-autoregressive moving average model
13	Spatio-temporal analysis of COVID-19 in India - a geostatistical approach
14	Epidemic size, trend and spatiotemporal mapping of SARS-CoV-2 using geographical information system in Alborz Province, Iran
15	Spatiotemporal dynamics and potential ecological drivers of acute respiratory infectious diseases: an example of scarlet fever in Sichuan Province
16	Association between satellite-detected tropospheric nitrogen dioxide and acute respiratory infections in children under age five in Senegal: spatiotemporal analysis

2.2.1 Eligibility criteria

Specific eligibility criteria were developed based on the main topic of the research question: “Machine learning models for infectious disease forecasting”. Studies must fulfill the specified inclusion criteria, which encompass the following conditions: 1) studies published between 2014 and 2024, 2) studies focusing on the prediction of ID cases, deaths, trends, transmission patterns, or outbreaks in the human population, and 3) studies focusing on the use of AI techniques for ID forecasting. Preprints, book chapters, conference presentations, reviews, opinions, commentaries, dissertations, and articles not written in English are excluded from the search process.

After the first stage of the search process, duplicate publications are removed, and a subsequent screening phase considers only research articles published from 2014 to 2024. In the second stage, applying the rest of the inclusion and exclusion criteria, papers are rejected based on abstracts, titles, and keywords. The third stage is selection, during which full-text readings are conducted and the decision to accept or reject the studies is taken. Thus, research studies are approved or rejected based on the criteria mentioned in Table 5.

Table 5: Inclusion and Exclusion criteria.

Inclusion criteria	Exclusion criteria
Studies published between 2014 and 2024	Preprints, book chapters, conference presentations, reviews, opinions, commentaries, dissertations, and articles not written in English.
Studies focusing on the prediction of ID cases, deaths, trends, transmission patterns, or outbreaks in the human population	
Studies focusing on the use of AI techniques for ID forecasting	

2.2.2 Quality Assessment

By evaluating the quality of the primary studies, it is guaranteed that the information discovered is relevant and adds value to the current research work. Every paper considered for the review features a prediction model for infectious diseases, utilizing either deep learning or machine learning. Addi-

tionally, the paper presents evidence of the model’s performance, including metrics such as accuracy, precision, sensitivity, or specificity.

2.3 Results

Two main topics are covered as of result of the literature review execution, 1) factors influencing infectious diseases, and 2) machine learning models for infectious disease forecasting.

2.3.1 Factors influencing infectious diseases

The prevalence of infectious diseases is influenced by a multitude of factors, which can be broadly categorized into environmental, biological and socioeconomic factors:

1. **Environmental Factors.** Such as temperature, humidity, and rainfall influence the prevalence of infectious diseases. These factors affect the survival, reproduction, and transmission dynamics of pathogens and their vectors. Understanding the influence of environmental factors on infectious diseases helps in predicting outbreaks and implementing effective public health interventions. Monitoring climate and weather patterns can inform strategies for disease prevention and control.
2. **Biological Factors.** Virulence, mode of transmission, and mutation rates of pathogens affect their spread and impact. These factors interact in complex ways to influence the dynamics of infectious diseases. For instance, a highly virulent pathogen that spreads easily through air can cause rapid and widespread outbreaks, especially in populations with low immunity. Similarly, the presence of efficient vectors in an environment can lead to high transmission rates of vector-borne diseases.
3. **Socioeconomic Factors.** Such as poverty significantly increase vulnerability to infectious diseases in several interconnected ways. Poor living conditions can lead to malnutrition, which weakens the immune system and increases susceptibility to infections. Limited access to healthcare exacerbates this by preventing timely treatment and reducing vaccination rates, further increasing disease prevalence and severity. Addressing these interconnected issues is crucial for reducing the overall burden of infectious diseases.

2.3.1.1 Land surface temperature

Temperature plays a significant role in the transmission dynamics of COVID-19. Research has shown that higher temperatures generally correlate with a reduction in the transmission risk of the virus, while lower temperatures tend to facilitate its spread. For instance, studies indicate that higher temperatures, particularly those exceeding 30°C, combined with moderate relative humidity levels of 45-60%, create conditions that are less favorable for the survival of the virus, thereby reducing the infection risk. Conversely, cold and damp environments have been identified as significantly increasing the transmission risk of COVID-19 (D. Wang et al., 2022; Zoran et al., 2023).

In regions with high temperatures and moderate humidity, the virus's ability to survive and remain infectious is considerably diminished. This can be attributed to the fact that elevated temperatures can accelerate the degradation of the virus on surfaces and in the air, making it less likely to infect individuals. Moreover, ultraviolet radiation, which is more intense during warmer months, further contributes to the inactivation of the virus. Consequently, the infection rates in such environments tend to be lower, providing a natural mitigation effect against infectious diseases. Conversely, colder temperatures present a more conducive environment for the virus to thrive. In cold climates, the virus remains stable and infectious for more extended periods, both on surfaces and in the air. Additionally, cold temperatures often coincide with higher humidity levels, particularly indoors, which can facilitate the survival and transmission of respiratory droplets carrying the virus. The persistence of the virus in these conditions leads to a higher likelihood of transmission, especially in enclosed and poorly ventilated spaces. (Mao et al., 2023).

The relationship between temperature and influenza transmission is complex, involving various meteorological factors that impact the virus's survival and infectivity. Influenza viruses thrive in colder temperatures, which is why influenza outbreaks are more common during the winter months. This seasonal pattern is observed globally, with temperate regions experiencing higher influenza activity during their respective cold seasons. One significant study that explores this relationship is the construction of the Combinatorial Judgment Classifier (CJC) model for predicting influenza outbreaks in Hong Kong. This model takes into account several meteorological variables, including mean temperature, maximum temperature, and minimum temperature, to forecast influenza trends. The study found that temperature variables play a critical role in predicting influenza outbreaks. For example, lower temperatures were strongly associated with increased influenza transmission, while higher temperatures were linked to reduced virus activity. This is consis-

tent with the general understanding that influenza viruses are more stable and infectious at lower temperatures (Z.-x. Wang et al., 2022). The impact of temperature on influenza transmission is also evident in the behavior of the virus under different temperature conditions. Influenza viruses are more stable and remain infectious for longer periods at lower temperatures. This increased stability enhances the virus’s ability to spread, particularly in indoor environments where people are more likely to congregate during colder weather. In contrast, higher temperatures reduce the stability and infectivity of influenza viruses. During warmer months, the virus’s ability to survive on surfaces and in the air is diminished, leading to lower transmission rates.

Temperature is a critical factor in the dynamics of infectious disease transmission. Higher temperatures and moderate humidity levels tend to reduce the virus’s survival, while colder and damp conditions enhance its persistence and spread. This understanding is essential for developing effective public health interventions and preparing for seasonal variations in infection rates, accentuating the importance of tailored strategies to combat the spread of such pathogens in different environmental contexts. By incorporating temperature data into predictive models, health authorities can better anticipate and respond to seasonal variations caused by pathogens, ultimately reducing the burden of infectious diseases.

2.3.1.2 Humidity

Humidity plays a crucial role in the transmission and survival of infectious diseases, particularly respiratory viruses like COVID-19 and influenza. Research indicates that both high and low humidity levels can impact the transmission dynamics of COVID-19, moderate humidity levels (45-60%) combined with higher temperatures ($>30^{\circ}\text{C}$) create less favorable conditions for the virus’s survival and transmission. This range of humidity can help to reduce the virus’s ability to remain infectious on surfaces and in the air. Conversely, environments with either very low or very high humidity levels can enhance the virus’s stability. Low humidity levels can dry out mucous membranes in the respiratory tract, making individuals more susceptible to infections, while high humidity levels can increase the longevity of respiratory droplets that carry the virus, thus facilitating transmission (D. Wang et al., 2022).

Similar to COVID-19, influenza virus transmission is also significantly affected by humidity. Low relative humidity, often associated with cold weather, creates an environment conducive to the survival and spread of

the influenza virus. This is because low humidity levels can help the virus remain airborne for longer periods, increasing the likelihood of inhalation by susceptible individuals (Z.-x. Wang et al., 2022). While high temperatures generally reduce virus viability, this effect is significantly influenced by the concurrent humidity levels.

2.3.1.3 Precipitation

Precipitation can affect the airborne transmission of viruses. Rain can remove particles, including viruses, from the air, thus reducing the likelihood of airborne transmission. However, heavy precipitation might lead to increased indoor crowding, which can elevate the risk of virus spread indoors (Mao et al., 2023).

Precipitation directly affects humidity levels. Increased rainfall leads to higher humidity, which, as discussed, can have varying effects on virus survival. High humidity levels generally cause respiratory droplets to settle more quickly, reducing airborne transmission. However, if the humidity becomes too high, it can create conditions that preserve the virus in droplets, maintaining its infectivity (D. Wang et al., 2022).

Precipitation can influence the stability of viruses on surfaces. Wet conditions can either wash away contaminants or create moist environments that preserve viruses. For instance, wet surfaces can facilitate the survival of influenza viruses, which thrive in moist, cool conditions. Zoran et al. (2023) suggests that precipitation, by maintaining surface moisture, can contribute to the persistence of viruses on various surfaces, potentially leading to higher contact transmission rates during rainy seasons.

Precipitation influences the transmission of infectious diseases through its impact on airborne and surface stability of viruses, changes in humidity, and human behavioral patterns. By incorporating precipitation data into predictive models, infectious disease outbreaks can be timely anticipated and managed.

2.3.1.4 Air pollution

Air pollution has a significant impact on the transmission and severity of infectious diseases, particularly respiratory infections such as COVID-19 and influenza. Air pollution can directly affect the respiratory system, making individuals more susceptible to infections. Pollutants such as particulate

matter (PM_{2.5} and PM₁₀), nitrogen dioxide (NO₂), and sulfur dioxide (SO₂) can cause inflammation and weaken the respiratory tract’s defenses, thereby facilitating viral infections (Zoran et al., 2023).

Long-term exposure to air pollution can lead to chronic respiratory and cardiovascular conditions, which are risk factors for severe outcomes in infectious diseases. For example, individuals with chronic obstructive pulmonary disease (COPD) or asthma, conditions often exacerbated by air pollution, are at higher risk for severe acute respiratory infections. Particulate matter can act as carrier for viruses, potentially increasing their airborne lifetime and transmission distance, facilitating their spread in the air (D. Wang et al., 2022).

Air pollution significantly impacts the transmission and severity of infectious diseases. Pollutants such as particulate matter can impair respiratory function and weaken immune defenses, making individuals more susceptible to infections. Long-term exposure to air pollution can lead to chronic respiratory conditions, which are risk factors for severe outcomes in ARIs (Li et al., 2022).

2.3.1.5 Socio-Economic Factors

Socio-economic conditions, including Gross Domestic Product (GDP), population density, and access to healthcare, are crucial drivers of ARIs. Higher GDP is associated with increased mobility and urbanization, which can enhance the spread of infectious diseases. Population density influences the transmission dynamics, with higher density areas experiencing more rapid spread of infections. During school holidays, the incidence of infections often decreases due to reduced contact among children, who are key transmitters of respiratory infections. Conversely, increased human mobility, such as through travel and migration, can facilitate the spread of pathogens across regions. Changes in land use, urbanization, and other environmental modifications can disrupt ecosystems and affect the transmission of infectious diseases. Urbanization can lead to higher population densities and increased contact rates, thereby facilitating the spread of ARIs (Li et al., 2022).

2.3.2 Forecasting models for infectious disease trends

Forecasting models for infectious diseases have become essential tools in managing and mitigating the impact of epidemics. These models leverage advancements in machine learning and deep learning to predict disease trends with remarkable accuracy, helping public health authorities implement timely interventions. Several forecasting models were found, providing an overview

of methodologies that incorporate both spatial and temporal data to enhance prediction accuracy. These models not only consider traditional epidemiological data but also integrate environmental, socio-economic, and demographic variables, offering a comprehensive approach to epidemic forecasting. This section presents case studies and methodologies that highlight the effectiveness of these models in different regions and under various conditions, emphasizing the critical role of advanced computational techniques in public health surveillance.

2.3.2.1 Construction of Influenza Early Warning Model Based on Combinatorial Judgment Classifier: A Case Study of Seasonal Influenza in Hong Kong

A Combinatorial Judgment Classifier (CJC) model is proposed by Z.-x. Wang et al. (2022) to classify epidemic trends and improve the accuracy of early warnings for influenza outbreaks. The study employs various machine learning methods, considering influenza data from Hong Kong, to provide a reliable early warning system.

The study utilized data from the National Influenza Surveillance Plan (2017 version) and the Hong Kong Department of Health. Data were collected from the 1st week of 2014 to the 33rd week of 2021. The characteristic variables for the model included atmospheric pressure, absolute maximum temperature, mean temperature, absolute minimum temperature, mean dew point temperature, the number of positive detections of seasonal influenza viruses, the positive percentage among all respiratory specimens, and the admission rates in public hospitals with a principal diagnosis of influenza.

Statistical analysis was performed using SPSS 26.0, and MATLAB was used for constructing the influenza early warning model. The data were divided into training and testing sets, and the prediction models included Discriminant Analysis, Support Vector Machine (SVM), and Ensemble Classifier. The final CJC model combined these classifiers to improve prediction performance.

The study focused on the meteorological and epidemic data from Hong Kong over an eight-year period. Variables were selected through one-way ANOVA analysis to improve model performance. These included meteorological data like atmospheric pressure and temperature, as well as epidemiological data like the number of positive influenza detections and hospital admission rates.

The study found that the CJC model had an accuracy of 96.47% for

predicting the influenza outbreak trend, with lower sensitivity and specificity change rates compared to other models, resulting in a precision of 90.0%, a recall of 71.05%, and a specificity of 99.16%. The CJC model showed high accuracy and stable prediction performance, outperforming other models like ARIMA. Thus, being able to provide reliable early warnings for influenza outbreaks, demonstrating its effectiveness in predicting both long-term and short-term epidemic trends.

The study also discussed the limitations, such as the need for regional data to improve accuracy and the potential for further improvements by incorporating more algorithms like Naïve Bayes and recurrent neural networks. Future research could involve data from different geographical regions to enhance the model’s applicability.

2.3.2.2 Spatial transmission network construction of influenza-like illness using dynamic Bayesian network and vector-autoregressive moving average model

Through the utilization of a Dynamic Bayesian Network (DBN) and a Vector Autoregressive Moving Average model (VARMA), Qiu et al. (2021) proposes a novel method to construct an influenza transmission network using spatio-temporal routes and to estimate the impact of influenza in one region on adjacent areas. The research focuses on Sichuan province, China, from 2010 to 2016.

The study used influenza-like illness (ILI) data from Sichuan province, encompassing 21 cities from 2010 to 2016. Sentinel surveillance data were collected, recording the number of ILI cases and total outpatient visits across various age groups. The ILI case numbers were calculated based on surveillance data and medical outpatient statistics.

The data showed a consistent seasonal pattern, with peaks in winter and spring. Four cities (Ziyang, Guangyuan, Yibin, and Panzhihua) were identified as key nodes in the transmission network, with significant impacts on neighboring cities. These cities are crucial transportation hubs, suggesting a strong link between influenza transmission and transportation routes.

The study introduced the concept of spatio-temporal routes, offering a novel approach to constructing spatial transmission networks without requiring detailed mobility data. The joint use of DBN and VARMA models proved effective in handling complex data structures and capturing the spatio-temporal correlations of ILI cases. The model can aid in timely and reliable predictions of influenza trends, helping public health authorities al-

locate resources effectively and implement targeted interventions.

2.3.2.3 Spatio-temporal modeling of COVID-19 prevalence and mortality using artificial neural network algorithms

Kianfar et al. (2022) investigate the relative importance of seventy five explanatory variables concerning COVID-19 prevalence and mortality using multilayer perceptron artificial neural network (ANN) topology. The study utilizes variable importance analysis (VIA) methods to identify the most influential variables over different periods, providing insights for public health policymakers. The explanatory variables, including demographic, environmental, social, economic, cultural, health, and public transportation, were compiled at the country level. COVID-19 Data were obtained from the World Health Organization (WHO) from March 2020 to February 2021, including new confirmed cases and deaths for all countries.

The study employed single-layer perceptron (SLP) neural networks with a non-linear sigmoid transfer function in the hidden layer and a linear function in the output layer. Through the use of a weighted Information Criterion (WIC) index and a Bayesian regularization method, the optimum number of neurons in the hidden layer were determined and a possible overfitting were addressed respectively.

The primary aim of using ANNs in this study is to predict future prevalence and mortality rates of COVID-19 based on historical data. The prevalence rate in the interquartile range (PR-IQR) was selected as the target for modeling COVID-19 prevalence, while the mortality rate (MR) was used for modeling COVID-19 deaths. In addition, the study employs ANNs to determine the relative importance of various explanatory variables influencing COVID-19 prevalence and mortality. This helps in identifying which factors are most significant in affecting disease outcomes, allowing policymakers to focus on critical areas for intervention. The performance of the proposed ANN models is evaluated using the normalized root mean square error interquartile index (RMSEIQR). This metric helps compare models on different scales, ensuring robustness in the results.

2.3.2.4 Computational forecasting methodology for acute respiratory infectious disease dynamics

By using a methodology that merges the predictions of a Feed Forward

Neural Network (FFNN), a Sum Of Sines (SoS) projection model, and a smoothed endemic channel calculation, González-Bandala et al. (2020) proposes a novel methodology combining computational models, machine learning, and internet search trends to predict ARI outbreaks accurately. The model aims to provide early warnings for epidemics, improving public health responses and resource allocation.

The study utilizes two primary data sources: 1) Weekly epidemiological data on ARIs reported by the Mexican Health Ministry, and 2) Google Trends data reflecting public search behaviors related to ARI symptoms and conditions.

The computational model proposed in this study integrates multiple advanced techniques to predict acute respiratory infection (ARI) outbreaks accurately. It comprises two primary components: a forecasting model and a projection model. The forecasting model utilizes a feedforward neural network (FFNN), which is trained to predict ARI cases based on the current week’s epidemiological data and public search behaviors captured through Google Trends. The FFNN analyzes the non-linear relationships between these inputs to generate short-term forecasts. Complementing this, the projection model employs a sum of sines (SoS) function, which fits historical ARI data to project future trends. This SoS model captures the cyclical nature of ARI outbreaks, providing a seasonal baseline for predictions. The final prediction is achieved by merging the outputs of the FFNN and the SoS model with smoothed endemic channels derived from historical data.

The FFNN model was tested over multiple winter seasons (2015-2019), showing high accuracy in predicting ARI cases one week in advance. The model’s performance improved with retraining every 13 weeks, reducing root mean square error (RMSE) and mean absolute percentage error (MAPE). The SoS projection model also demonstrated strong performance, accurately fitting the historical ARI data and providing reliable projections. The merge prediction model, which combines FFNN and SoS outputs with endemic channels, showed reduced RMSE, RMSPE, and MAPE, and higher correlation coefficients compared to individual models. This combined approach enhances prediction accuracy and reliability.

The results of the study demonstrated the effectiveness of the proposed computational model in predicting acute respiratory infection (ARI) outbreaks with high accuracy. The model’s performance was evaluated using two key metrics: Root Mean Square Percentage Error (RMSPE) and Mean Absolute Percentage Error (MAPE). For the forecasting model (FFNN), the RMSPE was significantly reduced with a retraining interval of every 13 weeks, indicating the model’s robustness in adapting to new data. The MAPE, which measures the average magnitude of errors in the predictions,

also showed substantial improvement. Specifically, the combined merge prediction model, which integrates the FFNN and SoS outputs with endemic channels, achieved a lower RMSPE and MAPE compared to individual models. This enhancement in prediction accuracy underscores the model’s ability to provide reliable forecasts, facilitating better public health planning and resource allocation. Quantitatively, the RMSPE (11.27%) and MAPE (30.90%) metrics indicated that the combined model outperformed traditional machine learning approaches such as support vector machines (SVM) and AdaBoost, as well as long short-term memory networks (LSTM), thereby validating its superior predictive capability.

2.3.2.5 A spatiotemporal machine learning approach to forecasting COVID-19 incidence at the county level in the USA

Lucas et al. (2023) introduce a novel approach using a spatiotemporal deep learning model, named COVID-LSTM, to forecast COVID-19 incidence at the county level in the United States. This model integrates spatial features derived from human movement and county connectedness data with temporal incidence data.

The dataset used in this study includes COVID-19 incidence data from Johns Hopkins University’s Center for Systems Science and Engineering, spanning from April 1, 2020, to February 20, 2021. The data consists of the cumulative number of confirmed COVID-19 infections per county per day.

Spatial features are derived from Facebook’s Social Connectedness Index (SCI) and Movement Range datasets. The SCI measures the connectedness between counties based on Facebook friendships, while the Movement Range dataset tracks the flow of the population within and between counties. Weather data, including average minimum and maximum temperatures, are also incorporated.

The COVID-LSTM model is an ensemble of 10 LSTM-based networks, each trained separately with randomly initialized parameters. The model architecture includes three hidden layers: two LSTM layers with 64 units each and a dense layer with 32 units. The final output is a single node representing the forecasted COVID-19 incidence. The model is trained for 15 epochs using the Adam optimizer with a learning rate of 0.001, and the ensemble prediction is the median of all constituent model predictions.

The primary AI model used in this study is the Long Short-Term Memory (LSTM) network, a type of recurrent neural network (RNN) known for its

ability to retain information over long sequences. The COVID-LSTM model captures temporal patterns in the data while incorporating spatial features to account for the heterogeneous spread of the virus. The ensemble approach, which combines multiple LSTM networks, is employed to reduce variability and improve the overall performance of the model.

The performance of the COVID-LSTM model is evaluated using two primary metrics: Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). These metrics are chosen to align with the standards used by the COVID-19 Forecast Hub, the primary reference for COVID-19 forecasting in the USA. The results demonstrate that COVID-LSTM outperforms the COVIDhub-ensemble model, which is used by the CDC for federal decision-making, over multiple forecast horizons. Specifically, the COVID-LSTM model is 11, 34, and 50 cases per county more accurate than the COVIDhub-ensemble for 2-, 3-, and 4-week forecast horizons, respectively.

2.3.2.6 Predicting the transmission trend of respiratory viruses in new regions via geospatial similarity learning

Traditional methods rely on prophase epidemiological data from the same region, which is not always available for new regions.. Zhao et al. (2023) propose a Contrastive learning-based Hierarchical Graph Convolutional Neural Network (CHGCN) to predict early-stage human-to-human transmitted respiratory infectious diseases (H-HRID) trends in new regions without prophase epidemiological data.

The data collection for this study focused on two major urban areas: San Francisco and New York. It involved gathering data on Points of Interest (POIs) and COVID-19 case rates to construct a comprehensive dataset for modeling. POIs included a variety of locations such as dining places, schools, and shopping centers, each characterized by attributes like longitude, latitude, height, and size. Additionally, the study utilized epidemiological data on COVID-19 case rates from official health sources to validate the model’s predictions. This detailed collection of geospatial and epidemiological data enabled the researchers to create accurate graph representations of each region, essential for training the Contrastive learning-based Hierarchical Graph Convolutional Neural Network (CHGCN) model.

The proposed model in this study is the Contrastive learning-based Hierarchical Graph Convolutional Neural Network (CHGCN), designed to predict early-stage trends of human-to-human transmitted respiratory infectious diseases (H-HRIDs) in new regions without relying on prior epidemiological

data. The CHGCN transforms the prediction task into a geospatial feature learning problem using graph neural networks. Each region is represented as a graph, with nodes corresponding to Points of Interest (POIs) and edges reflecting road network distances. Additionally, a region-level graph captures interactions between neighboring regions. The model employs a triplet contrastive learning approach to train itself, where the objective is to minimize the embedding difference between similar regions and maximize the difference between dissimilar ones. Through this structure, the CHGCN model effectively captures both local and global geospatial relationships, enabling accurate predictions of disease transmission trends.

The training of the CHGCN involves several critical steps to ensure accurate and robust predictions of early-stage respiratory infectious disease trends. The process begins with the creation of graph representations for each region, where nodes represent Points of Interest (POIs) characterized by features such as category, geographical coordinates, height, and size, and edges represent the road network distances between these POIs. The model then leverages the Relaxed Graph Edit Distance (RGED) algorithm to generate training data, creating triplets of graphs that consist of a target graph, a similar graph, and a dissimilar graph. These triplets are used in the triplet contrastive learning approach to train the model, which aims to reduce the embedding difference between similar regions while maximizing the difference between dissimilar ones. The Hierarchical Graph Convolutional Network (HGCN) within the model aggregates node features and updates graph-level embeddings using an attention mechanism that accounts for the significance of neighboring regions. This multi-layered training process, incorporating both local and regional geospatial data, equips the CHGCN model with the ability to make precise predictions about disease transmission trends in new regions.

The CHGCN model was evaluated using metrics such as Accuracy, Mean Reciprocal Rank (MRR), and Hit Rate at k (HR@ k), the model achieved an accuracy of 74.0% in predicting COVID-19 trends in San Francisco based on data from New York, highlighting its ability to generalize across different geographical locations. Additionally, the model showed a performance in MRR and HR@ k metrics, with 28.7% and 71.3% respectively. These results underscore the model’s robustness and reliability in capturing the spatiotemporal dynamics of infectious disease transmission.

2.3.2.7 Deep evolutionary fusion neural network: a new prediction standard for infectious disease incidence rates

Through the use of evolutionary principles, Yao et al. (2024) propose a Deep Evolutionary Fusion Neural Network (DEFNN) model that integrates multiple prediction methods and data sources to improve the accuracy and generalizability of infectious disease predictions.

Seven infectious disease datasets are selected, including four national datasets (hepatitis C, hepatitis B, tuberculosis, and brucellosis) and three regional datasets from Chongqing (hepatitis C-R, hepatitis B-R, and varicella-R). These datasets are obtained from the National Health Committee and the Chongqing Health Commission. Additionally, population data is sourced from China’s annual statistical yearbooks, and geographic data from the National Basic Geographic Information Database. In addition, Internet data are source from the Baidu index platform, focusing on the search and click frequency of key texts related to infectious diseases.

The DEFNN model introduced in the paper combines multiple prediction methods to accurately forecast infectious disease incidence rates. The DEFNN integrates linear models like the Seasonal Autoregressive Integrated Moving Average (SARIMA) with nonlinear models such as the Nonlinear Autoregressive Neural Network (NAR) and Long Short-Term Memory (LSTM) networks. The fusion approach involves using the residuals from linear models to train the nonlinear models, thereby enhancing the prediction accuracy. Central to the model is the Coronavirus Herd Immunity Optimizer (CHIO), a metaheuristic algorithm inspired by herd immunity principles, which optimizes the hyperparameters of the neural networks. This optimization process ensures the model effectively captures both linear and nonlinear trends in the data. Additionally, the DEFNN incorporates multisource data, including internet search trends to correct for traditional data lags and provide a more accurate, real-time prediction of disease trends.

The optimal hyperparameters for each model are determined using CHIO. The final DEFNN model includes SA-NAR and SA-LSTM components, tested on both national and regional datasets for validation. Overall, the DEFNN models improved indicators such as Mean Square Error (MSE), Mean Absolute Error (MAE), Root Mean Square Error (0.5233), and R-square (R2), reflecting better fitting and trend prediction capabilities in forecasting COVID-19 incidence at the county level.

2.3.2.8 Oriented transformer for infectious disease case prediction

By leveraging multiple representation fusion (MRF) methods to capture diverse temporal dependencies within disease case data, Z. Wang et al. (2023) introduce a method for predicting infectious disease cases using a model called Oriented Transformer (ORIT). This model aims to address the challenges of predicting infectious diseases, which include the variability of incubation periods and the complex dynamics of disease progression.

The ORIT model combines multi-orientation context vectors to capture multi-dimensional temporal relationships within disease case data. It utilizes a Multi-head Oriented Attention Unit (MOAU) to generate attention representations from various orientations, enabling the model to capture diverse patterns such as short-term trends, long-term trends, and rare events. The temporal fusion layer then aggregates these oriented attention representations to produce accurate predictions. The model is evaluated using two real-world datasets of infectious diseases: hand, foot, and mouth disease (HFMD) and hepatitis beta virus (HBV). Several traditional and advanced prediction methods are compared, including Auto-Regression (AR), Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Encoder-Decoder (ED), Convolutional Neural Network (CNN1D), and CNN-RNN hybrids, as well as variations of these methods enhanced with attention mechanisms.

The model's performance was analyzed based on parameters such as the number of time slices (B) and the look-back window size (T). Optimal values for these parameters were found for each dataset. The study showed that combining multiple orientations of attention significantly improved prediction accuracy. Comparative evaluations reveal that the ORIT model outperforms the aforementioned models across metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and the coefficient of determination (R^2).

Several key areas are outlined to further enhance the effectiveness and applicability of the ORIT model. Firstly, the integration of automated deep learning (ADL) strategies is proposed to streamline the process of hyperparameter optimization, potentially improving model accuracy and efficiency. Secondly, incorporating explainable artificial intelligence (XAI) techniques will be explored to enhance the interpretability and trustworthiness of the model's predictions. Thirdly, the inclusion of relevant exogenous features, such as environmental and socio-economic factors, is suggested to improve the model's robustness and adaptability to various scenarios. Lastly, the implementation of the ORIT model in online monitoring systems is envisioned to provide real-time predictive capabilities, enabling prompt and effective responses to infectious disease outbreaks.

2.3.2.9 Interpretable spatial identity neural network-based epidemic prediction

Based on fully connected neural networks and spatial identity matrix for predicting the number of epidemic infections, Luo et al. (2023) introduce the Interpretable Spatial Identity (ISID) neural network, which aims to predict infectious disease numbers at the regional weekly level using a simplified model structure that maintains high interpretability through post-hoc explanations using SHapley Additive exPlanations (SHAP).

Two primary datasets were employed: the US-Regions dataset and the Japan-Prefectures dataset. The US-Regions dataset, sourced from the United States Department of Health and Human Services, includes weekly influenza infection numbers across 10 regions from 2002 to 2017, covering 785 timeslots. The Japan-Prefectures dataset, obtained from the Japan Infectious Diseases Weekly Report, records weekly influenza infection numbers across 47 prefectures from August 2012 to March 2019, spanning 348 timeslots.

The model is designed to predict infectious disease incidence at the regional weekly level, focusing on balancing simplicity, accuracy, and interpretability. The model features a streamlined architecture consisting of three main components: an Embedding Multi-Layer Perceptron (MLP) layer, a Spatial Identity Matrix, and Residual Learning MLP layers. The Embedding MLP layer transforms input data into an embedded format suitable for further processing. The Spatial Identity Matrix is a learnable matrix that captures spatial relationships between different regions, enabling the model to understand and predict how diseases spread geographically. The Residual Learning MLP layers, comprising multiple fully connected layers with residual connections, enhance the model’s ability to learn complex patterns while avoiding overfitting. The ISID model uses a sliding window approach to make predictions based on historical data, and its predictions are explained post-hoc using SHapley Additive exPlanations (SHAP).

By using the Japan-Prefectures dataset and the US-Regions dataset, the model was compared with several traditional and deep learning models, including AR, ARMA, VAR, GAR, RNN, ATTRNN, DCRNN, LSTNet, STGCN, and Cola-GNN. Key metrics used for evaluation were Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Pearson Correlation Coefficient (PCC). For the Japan-Prefectures dataset, the ISID model achieved a Root Mean Square Error (RMSE) of 1622.780, a Mean Absolute Error (MAE) of 577.497, and a Pearson Correlation Coefficient (PCC) of 0.765 for a 3-week prediction horizon. Similarly, for the US-Regions dataset, the model achieved an RMSE of 947.311, an MAE of 486.672, and a PCC

of 0.862 for the same prediction horizon. These results indicate that the ISID model provides accurate predictions maintaining its performance across different regions and datasets.

2.3.2.10 LSTM-based recurrent neural network provides effective short term flu forecasting

By using a Long Short-Term Memory (LSTM) neural network, Amendolara et al. (2023) propose an approach to forecast short-term influenza-like illness (ILI) trends by leveraging historical ILI data, climate data, and population information. The model aims to accurately predict the weekly incidence of ILI, thereby enhancing preparedness and response strategies for influenza outbreaks focusing on offering a robust tool for real-time flu forecasting.

The study utilized a dataset spanning from week 40 of the 2003-2004 flu season to week 21 of the 2018-2019 season. This dataset includes weekly influenza-like illness (ILI) percentages and total patient visits recorded by the Centers for Disease Control and Prevention (CDC), along with detailed flu strain subtype information. In addition to the epidemiological data, the study incorporated climate data from the National Center for Environmental Information (NCEI), which includes temperature and precipitation measurements. Population data was sourced from the United States Census Bureau.

The architecture of the model included key predictors such as temperature, precipitation, wind speed, population size, vaccination rate, and vaccination efficacy. These predictors were selected based on their relevance to flu trends and their availability in the dataset. The LSTM network was designed to capture the temporal dependencies in the data, allowing it to learn patterns from historical ILI trends and climatic variables.

The model's performance was evaluated using Mean Absolute Error (MAE), achieving a +1 week prediction MAE of 0.1973. Key findings highlighted that temperature was the strongest predictor of ILI rates, with precipitation also enhancing the model's predictive capability when included. The model's optimal performance was observed with a four-week time lag, providing an optimal balance between prediction accuracy and data relevance.

2.3.2.11 CNN-LSTM deep learning based forecasting model for COVID-19 infection cases in Nigeria, South Africa and Botswana

Muhammad et al. (2022) introduce a hybrid deep learning algorithm combining Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, referred to as CNN-LSTM, to predict COVID-19 infection cases in Nigeria, South Africa, and Botswana. The proposed model combines the feature extraction capabilities of CNN with the temporal sequence learning abilities of LSTM. The CNN acts as an encoder to learn features from the input data, while the LSTM serves as a decoder to model short- and long-term temporal relationships.

The data used in this study was sourced from the public dataset provided by "Our World in Data," a project of the Global Change Data Lab. This comprehensive dataset includes 67 features related to COVID-19. From the available features, nine were selected for building the forecasting models: date, total cases, new cases, total deaths, new deaths, people vaccinated per hundred, people fully vaccinated per hundred, new deaths per million, and new cases per million. Additional features, such as moving averages for new cases and new deaths, were derived to account for sudden peaks and anomalies.

The model was evaluated using four key metrics: Accuracy, Mean Square Error (MSE), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE), outperforming both the individual CNN and LSTM models across all evaluation metrics. For Nigeria, the CNN-LSTM model achieved an accuracy of 98.30%, with MSE, MAE, and RMSE values of 0.00111, 0.00265, and 0.00265, respectively. Similar results were observed for South Africa and Botswana, where the CNN-LSTM model also demonstrated superior performance with accuracies of 97.60% and 97.74%, and correspondingly low error metrics.

2.3.2.12 Artificial Neural Network Prediction of COVID-19 Daily Infection Count

Despite advancements in vaccine development and treatment, COVID-19 remains a threat, especially with the potential emergence of new, more virulent variants. Accurate data collection and model assimilation are crucial for monitoring and responding to the pandemic. However, confirmed case counts reported by health agencies often represent only a fraction of the true infection count, complicating efforts to understand and mitigate the spread of the virus (Jiang et al., 2024).

This study aims to address the discrepancy between reported and true infection counts by employing an artificial neural network to uncover the relationship among daily confirmed case counts, testing data, population

statistics, and the actual daily case count. The ANN is trained using data from periods when death counts and infection fatality ratios (IFR) were more reliable, employing a backcasting technique to estimate the true infection count from death statistics and serological surveys.

The data collection and processing phase for the study involved several critical steps to ensure the accuracy and reliability of the input data used to train the artificial neural network (ANN). Data was sourced from the Johns Hopkins University Coronavirus Resource Center, including daily confirmed case counts and death counts, as well as additional information on population density and testing data. Serological survey data was also used to estimate the Infection Fatality Ratio (IFR). The preprocessing phase included correcting human errors, such as adjusting for artificial spikes caused by reporting backlogs, and removing weekend and holiday effects by averaging and interpolating the data. Smoothing techniques, specifically LOESS regression, were applied to reduce data noise, and exponential functions were used to address negative fluctuations in the initial phases of the data.

A feed-forward artificial neural network (FFNN) was employed to predict the true daily infection count of COVID-19, leveraging the nonlinear relationships between confirmed case counts, testing data, population density, and actual infection rates. The neural network architecture consisted of six layers, with sigmoid activation functions used to introduce nonlinearity and capture complex patterns in the data. The network also incorporated L2 regularization to prevent overfitting and ensure the model generalized well to unseen data. A key aspect of the training process involved data normalization and transformation to centralize the distribution of the training data, improving the efficiency and performance of the neural network. Additionally, a physics-informed neural network approach was adopted, introducing biology-informed regularization principles to further refine the model’s predictions. This approach ensured that the model’s predictions adhered to logical biological constraints, such as the true case count always being higher than the confirmed case count and having a positive relationship with testing volume.

Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) were used to assess the model’s performance, though the exact values for these metrics were not specified in the study. Nevertheless, the study demonstrated that the FFNN effectively predicted the true daily infection counts of COVID-19, showcasing accurate in-sample validation and successful out-of-sample forecasting from March 1, 2022, to July 1, 2022. The model indicated significantly higher true infection counts compared to reported confirmed cases, especially in areas with lower testing rates, highlighting the critical need for robust testing.

2.3.2.13 Forecasting the spread of COVID-19 using deep learning and Big Data Analytics methods

Based on the need for all-inclusive studies that capitalize the spread of the COVID-19 pandemic on the entire regions on the African continent, Kiganda and Akcayol (2023) conducted a comprehensive investigation and analysis to forecast COVID-19 cases and identify the most critical countries in terms of the COVID-19 pandemic across all five major African regions. The study seeks to address the gap in inclusive research for the African continent by leveraging both statistical and deep learning models, including ARIMA, LSTM, and Prophet, to predict the spread of COVID-19 using confirmed cumulative case data and evaluating model performance with multiple metrics.

COVID-19 data were gathered from a humanitarian data exchange, which was categorized by the five major regions of Africa: Northern, Southern, Eastern, Western, and Central Africa. The dataset included cumulative COVID-19 positive cases. Then, the study employed three distinct models to forecast COVID-19 cases across the five major African regions: the autoregressive integrated moving average (ARIMA) model with a seasonal perspective, a long short-term memory (LSTM) network, and the Prophet model.

The study evaluated the performance of the forecasting models using seven key metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), Symmetric Mean Absolute Percentage Error (SMAPE), Peak Signal-to-Noise Ratio (PSNR), Normalized Root Mean Squared Error (NRMSE), and the R2 score. These metrics provided a comprehensive assessment of each model's accuracy, precision, and ability to handle variability in the data. The results revealed that the LSTM model consistently outperformed the ARIMA and Prophet models across all regions. For instance, in Northern Africa, the LSTM model achieved an RMSE of 13.4 and an R2 score of 0.92, indicating a closer match to the actual data trends. In Central Africa, the LSTM model showed an RMSE of 14.5 and an R2 score of 0.89, while in Southern Africa, it achieved an RMSE of 12.3 and an R2 score of 0.95. Similar superior performance was observed in Western Africa, with an RMSE of 11.7 and an R2 score of 0.93, and in Eastern Africa, with an RMSE of 13.1 and an R2 score of 0.91. The study concluded that the LSTM model is highly effective in capturing the complex temporal dynamics of COVID-19 spread, making it the most reliable tool for forecasting future cases and identifying the most vulnerable countries in Africa.

Since only cumulative cases were employed for this study, future work should focus on integrating additional data sources, such as demographic information, to further refine the forecasts and enhance their applicability in managing the pandemic.

2.3.2.14 Recurrent Neural Network Ensemble, a New Instrument for the Prediction of Infectious Diseases

Puleio (2021) aim to explore the use of an ensemble of recurrent neural networks (RNNs) to predict the trends of epidemic curves, specifically focusing on influenza. The research investigates the influence of different neural network architectures over the forecasted seasons, and the number of training seasons required to achieve accurate predictions.

The study employed a public database comprising seasonal influenza epidemic data collected by the Italian National Sentinel Influenza Surveillance System (InfluNet). The data spans from the 2003/2004 to the 2017/2018 influenza seasons, totaling 15 seasons. The dataset includes weekly reports from family doctors and pediatricians who document influenza-like illness (ILI) cases.

The neural networks employed are time-delay neural networks (TDNN), which have a simple feed-forward architecture with two hidden layers. Two architectures were tested: one with two neurons per hidden layer and another with four neurons in the first layer and two in the second. The study examined the impact of using different time delays (5 and 10 weeks) on the prediction accuracy. Finally, the ensemble consists of multiple neural networks trained independently, with ensemble sizes ranging from 5 to 15 networks.

The results indicated that the ensemble with a time delay of 10 weeks and an architecture of four neurons in the first layer and two in the second achieved the best prediction accuracy, with an R^2 value of 0.98, closely matching the real data. The second study demonstrated a strong correlation between the size of the training set and prediction quality. As the number of training seasons decreased, the accuracy of the predictions also declined. Ensembles with more neural networks (10 and 15) provided more stable and reliable results, highlighting the importance of sufficient training data for accurate predictions.

The research demonstrates that even simple neural network architectures can achieve high prediction accuracy when organized in ensembles. Future work will focus on testing more complex epidemics with additional input variables, such as vaccination rates and government countermeasures, to enhance

prediction performance and applicability.

2.3.2.15 Spatio-temporal prediction of the COVID-19 pandemic in US counties: modeling with a deep LSTM neural network

Nikparvar et al. (2021) propose a multi-variate Long Short-term Memory (LSTM) recurrent neural network to predict COVID-19 infection cases and deaths 1 to 4 weeks ahead at the county level in the US, providing fine-grained predictions of COVID-19 infection cases.

The data used in this study includes confirmed COVID-19 cases and deaths obtained from the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University, spanning 33 weeks from January 26 to September 12, 2020. Daily foot traffic patterns and points of interest (POI) data were collected from SafeGraph’s Places Schema dataset. The data was pre-processed to correct inaccuracies, such as instances where cumulative counts decreased, which were replaced with the previous day’s value. Weekly sums of confirmed cases and deaths were calculated according to the Morbidity and Mortality Weekly Report (MMWR) weeks used by the CDC.

A multi-variate and multi-time series LSTM (MTS-LSTM) network is proposed to forecast COVID-19 cases, deaths, and mobility across different US counties using a single model. The model is trained on multiple time series simultaneously, capturing the temporal dynamics and spatial heterogeneity of the pandemic. The use of LSTM layers allows the model to memorize long-term dependencies, making it suitable for epidemic forecasting.

The study’s results demonstrated that the multi-variate Long Short-Term Memory (MTS-LSTM) model effectively predicted COVID-19 cases and deaths at the county level, outperforming traditional models in several aspects. The key findings revealed that incorporating mobility data significantly improved the model’s predictive performance. The MTS-LSTM model was evaluated using metrics such as bias, variance, Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and the R2 score. The results showed that the MTS-LSTM model achieved lower MSE and RMSE values, indicating higher accuracy and precision in its predictions. Additionally, the model demonstrated a higher R2 score, reflecting a better fit to the observed data compared to traditional models. The study highlighted that the MTS-LSTM model could efficiently replace multiple specialized models with a single, more effective model, reducing computational complexity while providing fine-grained and reliable predictions of COVID-19 dynamics. This robust performance underscores the model’s potential as a valuable tool for public health decision-making

and pandemic management.

The results indicate that increasing the number of counties involved in training generally improves the accuracy and consistency of the predictions. Specifically, experiments showed that using larger fractions of counties for training reduced the Root Mean Squared Error (RMSE) and variance of the predictions, particularly for new infection cases. For example, when all counties were used, the model with mobility as a predictor demonstrated lower RMSE and variance compared to Model which did not include mobility data. This pattern was more consistent for death predictions, which were less noisy than new case data. The findings suggest that incorporating data from a broader geographic scope enhances the model's ability to generalize and predict COVID-19 trends across different regions, thereby providing more reliable and robust forecasts.

3 Research Proposal

We propose a machine learning model for predicting spatiotemporal trends of infectious diseases in Mexico by leveraging meteorological and mortality data. Recognizing the ongoing public health challenge posed by infectious diseases such as influenza and COVID-19, this project seeks to enhance forecasting accuracy through advanced machine learning and deep learning techniques. By integrating diverse data sources, including environmental factors and population statistics, the model will provide precise predictions of disease outbreaks, enabling timely and effective resource allocation and public health interventions. The proposal outlines a comprehensive methodology, from data collection and cleaning to model training and deployment, ensuring a robust and practical solution for infectious disease forecasting in Mexico.

3.1 Methodology

The methodology section of this research proposal outlines a systematic approach to developing a machine learning model for forecasting infectious disease trends in Mexico. To achieve this, we will adhere to the Knowledge Discovery in Databases (KDD) process, which systematically extracts valuable insights from complex datasets. The KDD process is a classic data science lifecycle consisting of multiple steps executed sequentially. Each step depends on the successful completion and results of the previous step. The foundational structure of this process was proposed by (Frawley et al., 1992) and has since been refined. By employing this structured approach, we aim to create a sophisticated tool capable of aiding public health decision-makers in effectively managing and allocating resources to mitigate the impact of infectious diseases. This adherence to KDD ensures a robust framework for generating reliable and actionable insights (Figure 2).

3.1.1 Data Selection

Data selection is a critical component of the Knowledge Discovery in Databases (KDD) process that involves identifying and retrieving relevant subsets of data for analysis. In large-scale datasets, not all variables, attributes, or instances may be pertinent to the analysis goals or hypotheses under investigation. Therefore, data selection focuses on extracting meaningful subsets of data that are conducive to effective pattern discovery, predictive modeling, or other data mining tasks. The primary objective of data selection is to reduce the dimensionality and computational complexity of the dataset while ensuring that the selected data subsets retain sufficient information for

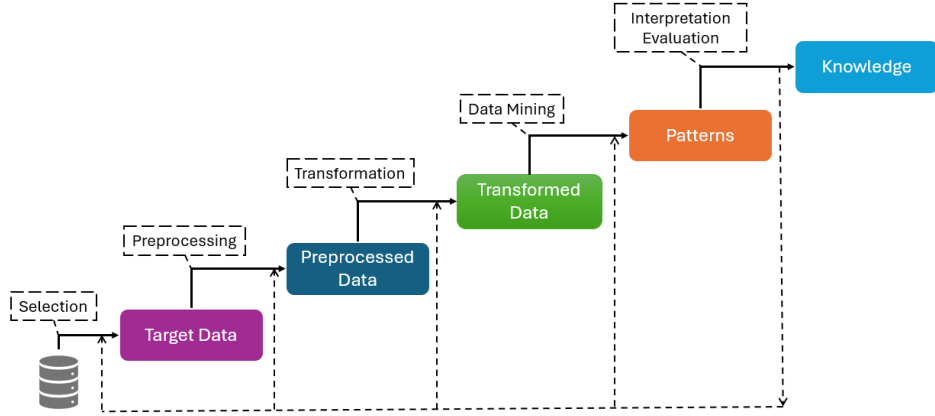


Figure 2: Phases of the KDD Process

accurate analysis and interpretation.

This step begins with identifying and defining the specific data requirements aligned with the research objectives, focusing on three primary domains: mortality records, population statistics, and meteorological data. Each of these data sources contributes unique and critical information that, when combined, offers a robust foundation for modeling the spatiotemporal dynamics of infectious diseases.

The process of data collection involves several key steps to ensure the integrity and relevance of the data. Initially, it requires obtaining access to the databases and ensuring that the data is comprehensive and up-to-date. Once the data is acquired, it is essential to verify its completeness and accuracy, ensuring that there are no significant gaps or inconsistencies that could affect the analysis. This step may involve cross-referencing multiple sources or datasets to validate the information.

In addition to gathering the data, it is crucial to understand the context and methodology used in its collection to ensure compatibility and relevance. For instance, the methods used to record mortality data and population statistics must be consistent over time to allow for accurate trend analysis. Similarly, the resolution and frequency of meteorological data must be suitable for the temporal and spatial scales of the study.

The data selection step is a comprehensive and meticulous process that lays the foundation for the entire research project. By systematically gathering and integrating high-quality data this phase ensures that the machine learning model has the robust and accurate input it needs to effectively predict infectious disease trends. This integrated approach not only enhances the model's predictive capabilities but also provides valuable insights that can

inform public health strategies and interventions, ultimately contributing to better management and mitigation of infectious disease outbreaks.

3.1.2 Data Pre-processing and cleansing

This step is an essential phase in the research methodology that involves refining raw data to enhance its quality, ensuring it is accurate, consistent, and ready for analysis. This process is crucial for improving the reliability of the machine learning model and involves several tasks such as handling missing values, detecting and addressing outliers, eliminating duplicates, and standardizing data formats.

3.1.2.1 Handling Missing Values

One of the primary tasks in data cleaning is managing missing values, which can arise due to various reasons such as incomplete data entry or loss of data during collection. Missing data can significantly impact the performance of a machine learning model if not properly handled. There are several strategies to address missing values:

- **Removal of Records:** In cases where the proportion of missing data is small, affected records can be removed to prevent bias.
- **Imputation:** For more significant gaps, imputation techniques are used. This involves replacing missing values with estimated ones, using methods such as mean, median, or mode imputation for numerical data, or the most frequent value for categorical data.

3.1.2.2 Detecting and Addressing Outliers

Outliers are extreme values that deviate significantly from other observations in the dataset and can distort statistical analyses and model training. Identifying outliers involves statistical methods and visualizations. Once identified, outliers can be addressed by removal, transformation or capping.

3.1.2.3 Eliminating Duplicates

Duplicate records can occur due to repeated data entry or merging datasets from multiple sources. These duplicates can skew the results and need to be identified and removed.

3.1.2.4 Data Standardiation

Ensuring consistency in data formats is critical for efficient data analysis. This involves: 1) Converting all date and time fields to a standard format, 2) ensuring that categorical variables use consistent labeling, such as using ‘Male’ and ‘Female’ consistently instead of mixing with ‘M’ and ‘F’, and 3) converting all numeric fields to a consistent unit of measure.

3.1.2.5 Dealing with Erroneous Data

Erroneous data, which includes incorrect or illogical values, can arise from data entry errors or issues during data collection. This type of data needs to be corrected or removed to ensure data integrity.

3.1.2.6 Ensuring Data Quality

Ensuring data quality is paramount for the success of the machine learning model and involves continuous validation and verification processes to maintain the integrity and accuracy of the dataset. This entails regular data profiling to identify and address inconsistencies, anomalies, and inaccuracies that could impact the analysis. Proper documentation of data sources, collection methods, and cleaning procedures is essential to ensure transparency, reproducibility, and accountability. By implementing robust validation rules and cross-referencing data with reliable sources, the integrity of the dataset is maintained, thereby enhancing the reliability and validity of the model’s predictions. Utilizing automated tools and scripts for data cleaning and validation further enhances efficiency, ensuring that the data remains accurate, consistent, and ready for high-quality analysis.

3.1.3 Data Transformation

This step focuses on transforming raw data into meaningful features that enhance the predictive power of the model. This process involves creating new features, modifying existing ones, and selecting the most relevant features to improve the model’s performance. The goal is to extract and highlight patterns from the data that are most pertinent to the predictive task, thereby enabling the model to learn effectively from the dataset. This step can be crucial for the overall success of the KDD process, since it is usually tight to the project.

One of the first tasks is to handle temporal data, especially when dealing with time series or sequential data. Lag features such as the number of

cases from previous weeks, are particularly useful in time-series forecasting, capturing the momentum and trends in disease spread. Spatial data also requires specific handling to extract meaningful features, this might include the distance to the nearest health facility, the density of healthcare resources, or regional mobility patterns. Spatial clustering techniques can be used to identify hotspots of disease incidence, which can then be used as features to enhance the model's spatial awareness.

Transforming raw numerical data into more informative features is another key aspect. For instance, meteorological data often needs to be converted from raw measurements to derived features that better represent their impact on disease dynamics. Similarly, temperature data can be processed to derive features such as the average, minimum, and maximum temperatures over specific periods, providing a more comprehensive picture of environmental conditions.

Categorical data requires different strategies to be transformed into numerical features that machine learning algorithms can process. Techniques such as one-hot encoding or label encoding are commonly used to convert categorical variables into binary or ordinal features. For example, different regions or districts can be encoded as separate binary features to capture regional variations in disease trends.

Feature scaling and normalization are also essential steps, since machine learning models often perform better when numerical features are on a similar scale. This helps in speeding up the convergence of gradient-based optimization algorithms and improves the overall performance of the model.

3.1.4 Data Mining

The Data Mining step involves applying advanced algorithms and statistical techniques to analyze data and uncover patterns, relationships, and trends. This step is central to the KDD process, as it transforms processed data into actionable insights. Data Mining includes various methods such as classification, clustering, regression, association rule learning, and anomaly detection. The goal is to identify hidden patterns and relationships within the data that can provide valuable insights, support decision-making, and predict future trends.

In this step, the most appropriate algorithm is chosen to achieve the research objectives effectively. This process involves evaluating multiple models to identify which one best captures the underlying patterns in the data and provides accurate predictions. The evaluation of these models is based on a range of performance metrics that reflect the accuracy and reliability of the predictions. Metrics such as Mean Absolute Error (MAE), Root Mean

Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE) are used to quantify the prediction errors for continuous variables. Additionally, cross-validation techniques are employed to assess the generalizability of the models, ensuring that they perform well on unseen data and are not overfitting to the training set.

3.1.5 Interpretation and Evaluation

The evaluation step is the final phase of the KDD process, where the data are prepared for interpretation and documentation. This step is crucial for ensuring that the patterns and insights discovered during the Data Mining phase are meaningful, valid, and actionable. Here, the discovered knowledge is assessed to determine its relevance and usefulness in the context of the analysis's specific goals. This involves evaluating the results for accuracy, reliability, and significance using various statistical and validation techniques. Additionally, this step includes interpreting the findings to understand their implications, ensuring they align with domain knowledge and are comprehensible to stakeholders. This phase often involves visualizing the results in a clear and comprehensible manner, facilitating effective communication and decision-making.

3.2 Data sources

The success of any research in computational forecasting relies on the quality and comprehensiveness of the datasets utilized. This section provides a detailed overview of the data sources employed to build the predictive models for the research proposal. The primary datasets include NASA's MODIS Land Surface Temperature and Emissivity data, MODIS Aerosol Product, Global Precipitation Measurement, mortality data from the National System of Statistical and Geographical Information (INEGI), and population data also provided by INEGI. These datasets offer critical insights into environmental, demographic, and epidemiological factors that influence the dynamics of infectious diseases. By integrating these diverse data sources, the research aims to develop robust machine learning models capable of accurately predicting the spatiotemporal trends of infectious diseases in Mexico. The preprocessing, aggregation, and alignment of these datasets ensure that they are ready for sophisticated analysis and model training, ultimately enhancing the precision and reliability of the forecasts.

3.2.1 MODIS Land Surface Temperature and Emissivity

NASA’s Moderate Resolution Imaging Spectroradiometer (MODIS) is designed to observe a wide range of Earth’s surface and atmospheric parameters, contributing to our understanding of global dynamics and processes. The MOD11 product provides Land Surface Temperature (LST) and emissivity data with a spatial resolution of 1 km at 1-day intervals. LST is a critical variable in climate modeling and environmental studies, it represents the temperature of the Earth’s surface. Emissivity, on the other hand, is a measure of the efficiency with which a surface emits thermal radiation.

Several infectious diseases are influenced by environmental factors, including temperature and humidity. Higher temperatures can increase the transmission rates of infectious diseases. Similarly, changes in surface temperature can affect the habitat suitability for vectors and pathogens. By integrating LST data with other epidemiological and demographic data, the proposal aims to develop a robust machine learning model that can predict the spatiotemporal trends of infectious diseases. The detailed LST measurements from MOD11 enable the model to account for microclimatic variations that might influence disease dynamics.

MOD11 data is freely available from NASA’s Earthdata platform and can be accessed through various tools and APIs. The data is provided in Hierarchical Data Format (HDF), which includes multiple layers and attributes necessary for comprehensive analysis. Once acquired, the data undergoes preprocessing steps to ensure its usability and accuracy. This includes georeferencing the data to align with other spatial datasets, and resampling to match the spatial resolution of other variables, resulting in an interpretable Climate Modeling Grid having 7,200 columns and 3,600 rows, which represent the entire globe with a spatial resolution of 0.05 degrees (Fig. 3). Finally, a temporal aggregation is performed to align with epidemiological data collection periods.

The integration of MOD11 data into the predictive model involves several steps. First, the LST and emissivity data are combined with other environmental variables, such as precipitation, to create a comprehensive environmental dataset. This dataset is then linked with epidemiological data, including disease incidence rates and demographic information, at various spatial and temporal scales.

Feature engineering plays a crucial role in this process, where raw temperature and emissivity values are transformed into features that can enhance the predictive power of the model. For instance, average, maximum, and minimum temperatures over specific periods can be calculated to capture seasonal trends. Anomalies or deviations from long-term averages can be

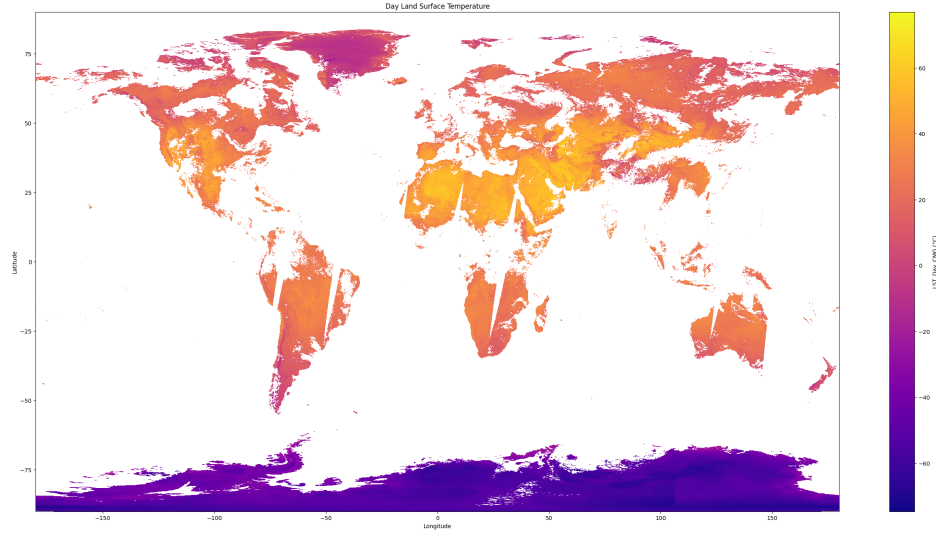


Figure 3: Terra MODIS daytime land surface temperature data from the MOD11C1 product across the globe, July 24, 2017.

identified to understand their correlation with disease outbreaks.

The use of MOD11 data in infectious disease forecasting offers several benefits. The high spatial and temporal resolution of the data provides detailed insights into the environmental conditions influencing disease dynamics. The ability to monitor temperature variations at fine scales helps in identifying local hotspots and understanding the microclimatic factors driving disease transmission.

Finally, Land Surface Temperature (LST) values are aggregated over relevant time periods and regions. This temporal aggregation allows the model to account for shifts in minimum, maximum, and mean temperatures. Additionally, spatial aggregation ensures that variations in LST across all municipalities in the Mexican territory are considered, enabling the detection of seasonal trends, anomalies, and deviations from average temperatures. Once both temporal and spatial aggregations are completed, the resulting dataset, as shown in Table 6, is stored for future processing.

3.2.2 MODIS Aerosol Product

The MODIS Aerosol Product (MOD08.D3) product specifically offers detailed measurements of aerosol optical thickness over the oceans globally and over the continents, which are essential for studying air quality, climate change, and their impacts on human health. The MOD08.D3 product provides aerosol optical depth (AOD) measurements at multiple wavelengths,

Table 6: Data structure for land surface temperature data integration.

Field name	Field description
CVEGEO	Municipality Unique Identifier
CVE_ENT	Entity number
CVE_MUN	Municipality number
TEMP_MIN_2003_01	Min. Land Surface Temperature (Year: 2003, Week 01)
TEMP_MED_2003_01	Mean Land Surface Temperature (Year: 2003, Week 01)
TEMP_MAX_2003_01	Max. Land Surface Temperature (Year: 2003, Week 01)
TEMP_MIN_2003_02	Min. Land Surface Temperature (Year: 2003, Week 02)
TEMP_MED_2003_02	Mean Land Surface Temperature (Year: 2003, Week 02)
TEMP_MAX_2003_02	Max. Land Surface Temperature (Year: 2003, Week 02)
⋮	⋮
TEMP_MIN_2022_52	Min. Land Surface Temperature (Year: 2003, Week 01)
TEMP_MED_2022_52	Mean Land Surface Temperature (Year: 2003, Week 01)
TEMP_MAX_2022_52	Max. Land Surface Temperature (Year: 2003, Week 01)

along with derived parameters such as aerosol size distribution. These measurements are available at varying spatial resolutions, typically 1 km to 10 km, and are aggregated into daily summaries.

AOD is a measure of the extinction of solar radiation by aerosol particles in the atmosphere. It provides an estimate of the columnar concentration of aerosols, which include dust, smoke, and industrial pollutants. Higher AOD values indicate more aerosols in the atmosphere, which can significantly affect air quality and visibility. Aerosols can have natural sources, such as dust storms and wildfires, as well as anthropogenic sources, such as industrial emissions and vehicle exhaust. Aerosol properties, are critical in understanding the relationship between air pollution and infectious diseases. Particulate matter in aerosols can serve as carriers for pathogens, enhancing

their transmission. Moreover, poor air quality can exacerbate respiratory conditions, making populations more susceptible to infections.

By integrating MOD08_D3 aerosol data with epidemiological and demographic data, this proposal aims to develop a machine learning model that can predict the spatiotemporal trends of infectious diseases. The detailed aerosol measurements from MOD08 enable the model to account for variations in air quality and their impact on disease dynamics. For instance, high AOD values can be correlated with increased incidence of respiratory infections, providing insights into how air pollution affects public health.

As well as MOD04, MOD08 data is freely available from NASA’s Earth-data platform and can be accessed through various tools and APIs. Once acquired, cloud contamination and other anomalies in the AOD measurements are identified and corrected using standard procedures, ensuring the reliability of the data. Once acquired, the data undergoes preprocessing steps to ensure its usability and accuracy. This includes georeferencing the data to align with other spatial datasets, and resampling to match the spatial resolution of other variables, resulting in an interpretable Climate Modeling Grid having a resolution of 360 columns and 180 rows with a spatial resolution of 1 degree, which represent the entire globe (Fig. 4). Finally, a temporal aggregation is performed to align with epidemiological data collection periods.

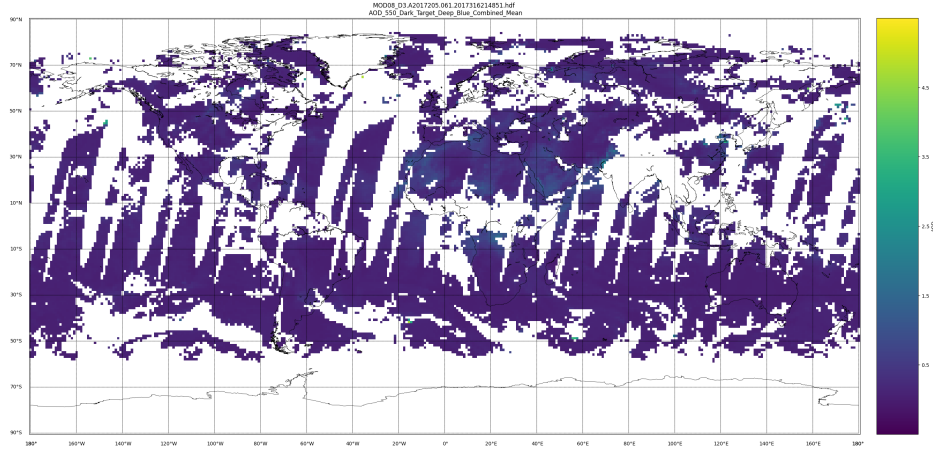


Figure 4: Terra Aerosol Optical Depth data from the MOD08_D3 product across the globe, July 24, 2017.

The integration of MOD08_D3 data into the predictive model involves several steps. First, the AOD and other aerosol properties are combined with environmental variables, such as temperature, to create a comprehensive environmental dataset. This dataset is then linked with epidemiological

data, including disease incidence rates and demographic information, at various spatial and temporal scales. The use of MOD08 data in infectious disease forecasting offers several benefits. The high spatial and temporal resolution of the data provides detailed insights into the environmental conditions influencing disease dynamics. The ability to monitor aerosol properties at fine scales helps in identifying local pollution hotspots and understanding the impact of air quality on disease transmission.

To match the temporal resolution of other integrated datasets, AOD values are aggregated over relevant time periods and regions. This temporal aggregation enables the model to account for shifts in air quality. Additionally, spatial aggregation ensures that variations in air quality across all municipalities in the Mexican territory are considered. Once both temporal and spatial aggregations are completed, the resulting dataset, as shown in Table 7, is stored for future processing.

Table 7: Data structure for air quality data integration.

Field name	Field description
CVEGEO	Municipality Unique Identifier
CVE_ENT	Entity number
CVE_MUN	Municipality number
AOD_2003.01	Mean Aerosol Optical Depth (Year: 2003, Week 01)
AOD_2003.02	Mean Aerosol Optical Depth (Year: 2003, Week 02)
⋮	⋮
AOD_2022.52	Mean Aerosol Optical Depth (Year: 2022, Week 52)

3.2.3 Global Precipitation Measurement

The Global Precipitation Measurement (GPM) mission aims to advance our understanding of global precipitation patterns combining precipitation observations from an international constellation of satellites. The GPM_3IMERGDF product is part of the Integrated Multi-satellite Retrievals for GPM (IMERG) suite, which combines data from GPM and other satellites to provide a comprehensive and continuous record of precipitation, delivering half-hourly precipitation estimates with a spatial resolution of 0.1 degrees (10 km).

The detailed precipitation measurements from GPM_3IMERGDF allow the model to incorporate the effects of rainfall on disease transmission dynamics. For example, high-resolution precipitation data can help identify periods of increased vector activity or areas at risk of waterborne disease outbreaks. Understanding these relationships is essential for developing targeted public health interventions and improving disease surveillance.

GPM data is freely available from NASA’s Precipitation Processing System (PPS) and can be accessed through various tools and APIs. The data is provided in NetCDF-4 (Network Common Data Form version 4), which is a set of software libraries and machine-independent data formats that support the creation, access, and sharing of scientific data. Once acquired, the data undergoes preprocessing steps to ensure its usability and accuracy. This includes georeferencing the data, and resampling to match the spatial resolution, resulting in an interpretable Climate Modeling Grid having 3,600 columns and 1,800 rows, which represent the entire globe (Fig. 5). Finally, a temporal aggregation is performed to align with epidemiological data collection periods.

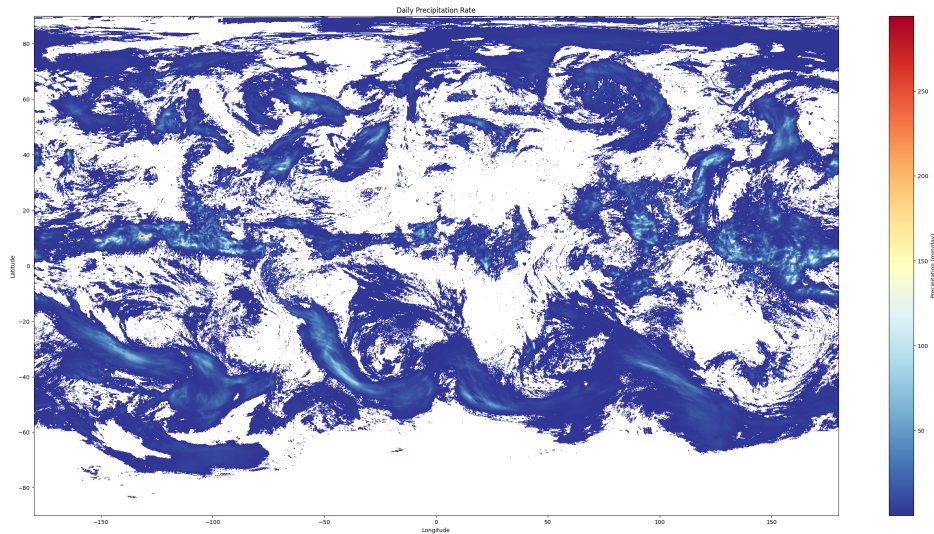


Figure 5: Daily accumulated precipitation data from the GPM_3IMERGDF product across the globe, July 24, 2017.

The use of GPM_3IMERGDF data in infectious disease forecasting offers several benefits. The high spatial and temporal resolution of the data provides detailed insights into the environmental conditions influencing disease dynamics. The ability to monitor precipitation patterns at fine scales helps in identifying local hotspots and understanding the impact of rainfall

on disease transmission.

To match the temporal resolution of other integrated datasets, precipitation rates are aggregated over relevant time periods and regions. This temporal aggregation enables the model to account for shifts in precipitation rates that may influence disease patterns. Additionally, spatial aggregation ensures that variations in precipitation rates across all municipalities in the Mexican territory are considered. Once both temporal and spatial aggregations are completed, the resulting dataset, as shown in Table 8, is stored for future processing.

Table 8: Data structure for precipitation data integration.

Field name	Field description
CVEGEO	Municipality Unique Identifier
CVE_ENT	Entity number
CVE.MUN	Municipality number
PRECIP_2003_01	Weekly accumulated precipitation (Year: 2003, Week 01)
PRECIP_2003_02	Weekly accumulated precipitation rate (Year: 2003, Week 02)
⋮	⋮
PRECIP_2022_52	Weekly precipitation rate (Year: 2022, Week 52)

3.2.4 Mortality Data

The National System of Statistical and Geographical Information (INEGI) is an autonomous public body responsible for collecting and disseminating information about Mexico in terms of territory, resources, population and economy. The mortality data from INEGI encompasses detailed records of registered deaths across the country. This dataset is vital for understanding the historical and spatial patterns of mortality, especially those related to infectious diseases.

The data is collected through a comprehensive network of civil registry offices and health institutions, ensuring that it covers the entire population. INEGI’s mortality data is typically available in annual reports and can be accessed through their online databases. This data provides a granular view of mortality trends, enabling researchers to analyze how different factors influ-

ence death rates and identify patterns that might be indicative of infectious disease outbreaks.

Mortality data is a critical component in infectious disease forecasting, as it provides direct evidence of the impact of diseases on the population. By analyzing trends in mortality data, researchers can identify outbreaks, assess the severity of disease spread, and evaluate the effectiveness of public health interventions. Integrating mortality data with other datasets, such as environmental and demographic information, allows for a more comprehensive understanding of the factors driving disease dynamics.

The first step in utilizing mortality data involves comprehensive data collection and meticulous preprocessing. The National System of Statistical and Geographical Information (INEGI) provides mortality data in various formats, including the widely-used Database File format (DBF), which can be accessed through their online portals. These files contain detailed records of mortality statistics, encompassing critical variables such as cause of death, date, location, and age. The accessibility and richness of this data make it a valuable resource for in-depth epidemiological analysis.

Once acquired, the data undergoes a rigorous preprocessing phase to ensure its usability, consistency, and accuracy. This preprocessing involves several key steps. First, data cleaning is performed to handle missing values, correct any inaccuracies, and standardize data formats. This step ensures that all records are complete and uniform, facilitating reliable analysis.

The next crucial step is data aggregation, where mortality data is summarized over specific time periods to match the temporal resolution of other integrated datasets, such as environmental and demographic data. For this proposal, mortality data is aggregated over daily time periods. This involves calculating the number of deaths per week for each spatial unit, which may include regions such as municipalities, states, or specific geographic coordinates.

Geospatial alignment is also an essential aspect of preprocessing. The mortality data must be georeferenced to ensure that it aligns accurately with other spatial datasets. This involves mapping the locations of deaths to precise geographic coordinates or administrative regions. Accurate georeferencing is critical for spatial analysis, enabling the identification of geographic patterns and clusters in mortality rates.

Temporal alignment is equally important, as it ensures that mortality data is synchronized with other temporal datasets. This step involves aligning the time periods of mortality data with those of environmental data (e.g., temperature, precipitation) and demographic data.

Feature engineering plays a pivotal role in enhancing the predictive power of the model. During this process, raw mortality data is transformed into

meaningful features that can be utilized by machine learning algorithms. They might also compute mortality rates by normalizing the number of deaths against the population size of each spatial unit, allowing for comparisons across different regions with varying population densities.

To effectively align mortality data with environmental and demographic data, a well-designed data structure is essential. Georeferenced fields are crucial as they facilitate seamless integration of datasets. Another key element to consider is the inclusion of weekly death counts grouped by region. For this research proposal, the data structure outlined in Table 9 will be employed to ensure accurate and efficient data alignment.

Table 9: Data structure for mortality data integration.

Field name	Field description
CVEGEO	Municipality Unique Identifier
CVE_ENT	Entity number
CVE_MUN	Municipality number
CAUSA_DEF	International Classification of Diseases Code
DEF_2003_01	Weekly deaths (Year: 2003, Week 01)
DEF_2003_02	Weekly deaths (Year: 2003, Week 02)
⋮	⋮
DEF_2022_52	Weekly deaths (Year: 2022, Week 52)

3.2.5 Population Data

The population data provided by INEGI includes detailed records on the demographic characteristics of the Mexican population, such as age, gender, occupation, educational attainment, income levels, and spatial distribution. This data is collected through nationwide censuses and surveys, ensuring comprehensive coverage and granularity. The data is available in various formats, including CSV, which can be accessed through the official online portals.

Population data is a critical element in infectious disease forecasting as it provides context for interpreting epidemiological data. Demographic factors,

such as population density, significantly influence the spread and impact of infectious diseases. For example, densely populated urban areas may experience faster disease transmission rates compared to sparsely populated rural regions.

Population data must be georeferenced to align with other spatial datasets, ensuring that population data can be integrated seamlessly with environmental and epidemiological datasets, which is critical for spatial modeling of disease trends. To match the temporal resolution of other integrated datasets, population data is aggregated over relevant time periods. Temporal aggregation allows the model to account for shifts in population dynamics that may influence disease patterns, the table 10 outlines the resulting data structure.

Table 10: Data structure for population data integration.

Field name	Field description
CVEGEO	Municipality Unique Identifier
CVE_ENT	Entity number
CVE_MUN	Municipality number
POB_2003_01	Estimated population per municipality (Year: 2003, Week 01)
POB_2003_02	Estimated population per municipality (Year: 2003, Week 02)
⋮	⋮
POB_2022_52	Estimated population per municipality (Year: 2022, Week 52)

By integrating population data with environmental and epidemiological information, the model can account for a wide range of variables that affect disease transmission and impact. The comprehensive approach outlined in this proposal underscores the importance of using high-quality statistical data to address complex public health challenges and improve disease surveillance and control strategies.

3.3 Machine Learning models

A machine learning model is a mathematical representation that encapsulates patterns and relationships within a dataset, enabling predictions, classifications, or decision-making without explicit programming. At its core, a machine learning model leverages algorithms that allow computers to learn

from data, identify patterns, and make decisions based on that learning. The concept of machine learning is rooted in the idea that systems can automatically improve their performance through experience, much like humans learn from observation and practice.

Machine learning models are broadly categorized into three types: 1) supervised learning, 2) unsupervised learning, and 3) reinforcement learning. In supervised learning, the model is trained on a labeled dataset, meaning the input data are paired with the correct output. This category includes tasks such as regression, where the goal is to predict continuous values, and classification, where the goal is to predict categorical labels. The model learns to map inputs to outputs by minimizing the difference between its predictions and the actual outcomes. Algorithms commonly used in supervised learning include linear regression, logistic regression, decision trees, support vector machines, and neural networks.

3.3.1 Support Vector Machine

A Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression tasks. Developed by Vladimir Vapnik and his colleagues in the 1990s, SVMs are based on the concept of finding a hyperplane that best separates the data into different classes. The primary goal of an SVM is to find the optimal hyperplane that maximizes the margin between the classes. The margin is defined as the distance between the hyperplane and the nearest data points from each class, which are known as support vectors. SVMs are known for their effectiveness in high-dimensional spaces and their robustness to overfitting, particularly in classification tasks.

While SVMs are traditionally used for classification tasks, they have been adapted for regression tasks as well (Support Vector Regression, or SVR), making them suitable for time series forecasting. In SVR, the algorithm tries to find a function that deviates from the actual observed values by a value no greater than a specified margin, while simultaneously being as flat as possible. SVMs have been widely used in financial time series forecasting, such as stock price prediction (Tay & Cao, 2001), energy load forecasting (Fan & Hyndman, 2011), and weather prediction (Danandeh Mehr et al., 2019).

Support Vector Machines (SVMs) have been successfully employed for weather prediction due to their ability to handle non-linear relationships and high-dimensional data. SVMs are used to predict daily, weekly, or monthly values. SVMs are a powerful tool for weather prediction, capable of handling the non-linear and high-dimensional nature of meteorological data. Through careful data preparation, kernel selection, and parameter tuning, they can

provide accurate forecasts for various weather parameters such as temperature, humidity, and precipitation rates.

3.3.2 Feedforward Neural Network

Feed Forward Neural Networks (FFNNs) are a fundamental type of artificial neural network used primarily for supervised learning tasks. A FFNN consists of three main types of layers: the input layer, which receives the input data; one or more hidden layers, where the data undergoes transformation through a series of weighted sums and activation functions; and the output layer, which produces the final prediction. Weights are numerical values that adjust the input data as it passes through each layer, while biases are additional parameters added to the weighted inputs to provide flexibility in the output. Activation functions, such as sigmoid, tanh, and ReLU (Rectified Linear Unit), introduce non-linearity into the model, enabling it to capture complex patterns. During forward propagation, data moves sequentially from the input layer to the output layer, generating predictions. The backpropagation algorithm is then used to minimize prediction errors by iteratively adjusting the weights and biases, based on the gradient of the loss function. This process of learning from data through forward and backward passes enables FFNNs to effectively model relationships in various types of data, making them a fundamental tool in machine learning and predictive modeling.

Feed Forward Neural Networks (FFNNs) are used for time series forecasting by transforming sequential data into a structured format suitable for these networks. This is typically achieved through feature engineering techniques, such as creating lag features that use past observations as inputs to predict future values, and employing a sliding window approach where a fixed-size window of historical data is used to forecast the next time step. By incorporating additional features like rolling statistics and seasonal indicators, FFNNs can capture complex patterns and non-linear relationships in the time series data. Despite their lack of inherent sequential processing, FFNNs can effectively learn from these engineered features to make accurate predictions, making them a versatile tool in various forecasting applications, such as weather forecasting and healthcare analytics (Khashei et al., 2009).

3.3.3 Convolutional Neural Network

A Convolutional Neural Network (CNN) is a type of deep learning model specifically designed to process and analyze data with grid-like topology, such as images. CNNs are characterized by their ability to automatically

and adaptively learn spatial hierarchies of features from input data through the use of convolutional layers. These networks have proven highly effective in tasks like image and video recognition, image classification, and even natural language processing.

CNNs adaptively learn spatial hierarchies of features through convolutional layers, which apply filters or kernels to the input data to produce feature maps that detect local patterns. The input data is processed through multiple layers to progressively extract higher-level features and generate predictions. Initially, the input layer receives raw data, which is then passed through convolutional layers where filters (kernels) slide over the input to create feature maps that detect local patterns. Activation functions, introduce non-linearity by zeroing out negative values. Pooling layers, such as max pooling, downsample the feature maps by selecting maximum values from patches, reducing spatial dimensions and computational load. Stacking multiple convolutional and pooling layers allows the network to learn increasingly complex features. The output from these layers is flattened and fed into fully connected layers, where each neuron is connected to all neurons in the previous layer, enabling high-level reasoning. The final output layer generates the prediction, often using a softmax function for classification tasks. The network is trained via backpropagation, where errors between predicted and actual outputs are minimized by adjusting weights through gradient descent, allowing the CNN to learn from the data and improve accuracy.

While CNNs are traditionally used for image data, they have been adapted for time series forecasting due to their ability to capture local patterns and temporal dependencies in sequential data (Borovykh et al., 2017). By treating time series data similarly to one-dimensional images, CNNs can learn from temporal and spatial features effectively.

3.3.4 Long Short-Term Memory Neural Network

Recurrent Neural Networks (RNNs) are a class of artificial neural networks designed for processing sequential data. Unlike traditional feedforward neural networks, RNNs have connections that form directed cycles, allowing information to persist. This capability makes RNNs particularly effective for tasks where context and sequential dependencies are critical, such as time series forecasting, natural language processing, and speech recognition. RNNs maintain a hidden state that captures information from previous time steps, enabling them to model temporal dependencies. They are trained using Backpropagation Through Time (BPTT) to update weights based on errors over multiple time steps. Key variants like Long Short-Term Memory (LSTM) networks incorporate gating mechanisms to control the flow of infor-

mation and mitigate issues like the vanishing gradient problem, making them particularly effective for tasks that require learning long-term dependencies.

LSTMs are capable of learning long-term dependencies in sequential data, making them highly effective for tasks that require context over extended periods. These kind of neural networks are used to predict future values in a time series by learning patterns and dependencies from historical data.

3.3.5 Graph Neural Network

A Graph Neural Network (GNN) is a type of neural network designed to work with graph-structured data. Graphs are data structures consisting of nodes (or vertices) and edges (connections between nodes), which represent relationships and interactions. GNNs learn a representation (embedding) for each node in the graph. These embeddings capture the node's own features as well as the structural information from its neighbors. The core mechanism of GNNs is the message passing process, where nodes exchange information with their neighbors. During each iteration (or layer) of the GNN, a node aggregates information from its neighbors and updates its own embedding based on this aggregated information.

GNNs can be used for time series forecasting, particularly in scenarios where the temporal data has underlying graph structures or where relational information between different time series needs to be leveraged. In a GNN applied to time series data, each node can represent a time step or a specific entity in the temporal dataset. Edges can capture the temporal or relational dependencies between the nodes. For example, an edge might connect entities that have a direct relationship, such as geographical proximity or similar behavior patterns.

3.4 Metrics

Several key metrics will be included to ensure robust and accurate predictions of the forecasting models. The Mean Absolute Error will measure the average magnitude of errors in the predictions, providing a straightforward interpretation of the model's accuracy. The Root Mean Squared Error will quantify the standard deviation of the prediction errors, offering insights into the model's precision and its sensitivity to larger errors. The Mean Absolute Percentage Error will express the prediction accuracy as a percentage, facilitating comparisons across different scales of data. Lastly, the R-squared metric will evaluate the proportion of variance in the dependent variable that is predictable from the independent variables, indicating the overall goodness of fit for the models. Together, these metrics will provide a comprehensive

assessment of each model's performance, guiding the selection of the most effective forecasting approach for predicting mortality trends related to infectious diseases.

3.4.1 Mean Absolute Error

Mean Absolute Error (MAE) is used to evaluate the accuracy of a forecasting model by measuring the average magnitude of errors between predicted and actual values. It is calculated by taking the absolute differences between the forecasted and observed values and then averaging these differences across all data points. The MAE provides a clear and straightforward interpretation of the model's accuracy, with lower values indicating better predictive performance. Its importance lies in its simplicity and ease of understanding, making it accessible to both technical and non-technical stakeholders. MAE is particularly useful because it treats all errors equally, providing an unbiased measure of prediction accuracy without being overly sensitive to outliers.

3.4.2 Root Mean Squared Error

Root Mean Squared Error (RMSE) is used to evaluate the accuracy of a forecasting model by measuring the square root of the average squared differences between predicted and actual values. It is calculated by taking the differences between the forecasted and observed values, squaring these differences, averaging them, and then taking the square root of this average. RMSE provides a comprehensive measure of prediction accuracy, with lower values indicating better model performance. Its importance lies in its sensitivity to larger errors, as the squaring process amplifies the impact of significant deviations.

3.4.3 Mean Absolute Percentage Error

Mean Absolute Percentage Error (MAPE) is a metric used to evaluate the accuracy of a forecasting model by expressing the prediction error as a percentage of the actual values. It is calculated by taking the absolute differences between the forecasted and observed values, dividing each by the actual value to convert them into percentages, and then averaging these percentages over the entire dataset. MAPE provides an intuitive understanding of the model's accuracy, as it directly relates the error to the scale of the data, making it easy to interpret and compare across different datasets.

3.4.4 R-squared

R-squared (R^2), also known as the coefficient of determination, is a statistical metric used to evaluate the goodness of fit of a regression model. It represents the proportion of the variance in the dependent variable that is predictable from the independent variables. R^2 values range from 0 to 1, where a value closer to 1 indicates that the model explains a large portion of the variance in the outcome variable, while a value closer to 0 indicates that the model explains very little of the variance. The importance of R-squared lies in its ability to provide a measure of how well the model captures the underlying trend in the data, offering insights into the explanatory power of the predictors.

4 Current progress

We have made significant progress in collecting and preparing data for the research. The data on land surface temperature (LST), precipitation, aerosol optical depth (AOD), mortality, and population have been gathered from a period of 20 years (from 2003 to 2022). Rigorous data cleaning processes were undertaken, this included handling missing values on the images corresponding to LST, precipitation, and AOD. In addition, erroneous data points were corrected or eliminated from the mortality registers.

All the data collected were aggregated into an epidemiological week period spanning from 2003 to 2022. This temporal aggregation aligns the data with standard epidemiological reporting practices, enabling consistent analysis over time. Additionally, LST, AOD, and precipitation data were spatially aggregated at the municipality level. This spatial aggregation ensures that the environmental data are precisely matched with the corresponding geographical regions, facilitating an accurate assessment of their impact on epidemiological trends within each municipality. This approach allows for a comprehensive analysis of both temporal and spatial factors influencing infectious disease patterns.

Mortality data were meticulously aggregated and summarized to enhance the analysis. This process resulted in the calculation of weekly deaths per municipality, categorized by International Classification of Diseases (ICD) codes. By organizing the mortality data in this manner, the research ensures a detailed and precise understanding of death trends and their potential correlations with environmental factors. The weekly aggregation allows for the identification of patterns and anomalies over time, while the categorization by ICD codes provides insight into specific causes of death, enabling a more nuanced analysis of public health trends at the municipal level.

Demographic information has been comprehensively included in the dataset, grouped by municipalities. This information encompasses total population, as well as geographical details such as latitude, longitude, and the area of each municipality. By integrating these demographic variables, the research provides a robust framework for analyzing how population dynamics and geographical characteristics influence epidemiological trends. The inclusion of population data allows for per capita calculations, which are crucial for understanding the true impact of diseases on different communities. Geographical coordinates enable spatial analysis, facilitating the study of environmental factors in relation to specific locations, while the area of each municipality helps contextualize population density and its potential effects on disease spread and mortality.

5 Work plan

The next phase of the research will focus on Data Transformation. Data analysis will be performed to uncover patterns, trends, and relationships within the data. Visualization techniques will be employed to illustrate distributions, correlations, and potential anomalies. Statistical analyses will summarize the key characteristics of the dataset. Feature Engineering will be applied to enhance the model's predictive power, with domain-specific feature selection to retain the most relevant variables. If necessary, dimensionality reduction techniques will be applied to manage multicollinearity and improve model efficiency.

Once the Data Transformation phase is completed, the resulting datasets will be employed to perform time series forecasting, specifically targeting the number of deaths related to infectious diseases. An initial approach will focus on the municipalities of a reduced number of entities, then, following an incremental process, the rest of the entities will be included into the proposed models to assess their performance. The models to be evaluated, as mentioned previously, include Support Vector Machine (SVM), Feedforward Neural Network (FFNN), Convolutional Neural Network (CNN), Long Short-Term Memory Neural Network (LSTM), and Graph Neural Network (GNN). Each model will undergo rigorous assessment using key evaluation metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R-squared (R^2). These metrics will provide comprehensive measures of the models' precision, error rates, and overall fit to the data. By systematically comparing the performance of each model, the most effective one for accurately predicting mortality trends will be identified. Additionally, an ongoing literature review will be conducted to gather further insights and benchmark the proposed models against existing state-of-the-art approaches, ensuring that our methodology remains at the forefront of infectious disease forecasting. Figure 6 details the activities to be accomplished on the following months, and Figure 7 details the activities and goals to achieve for this research proposal.

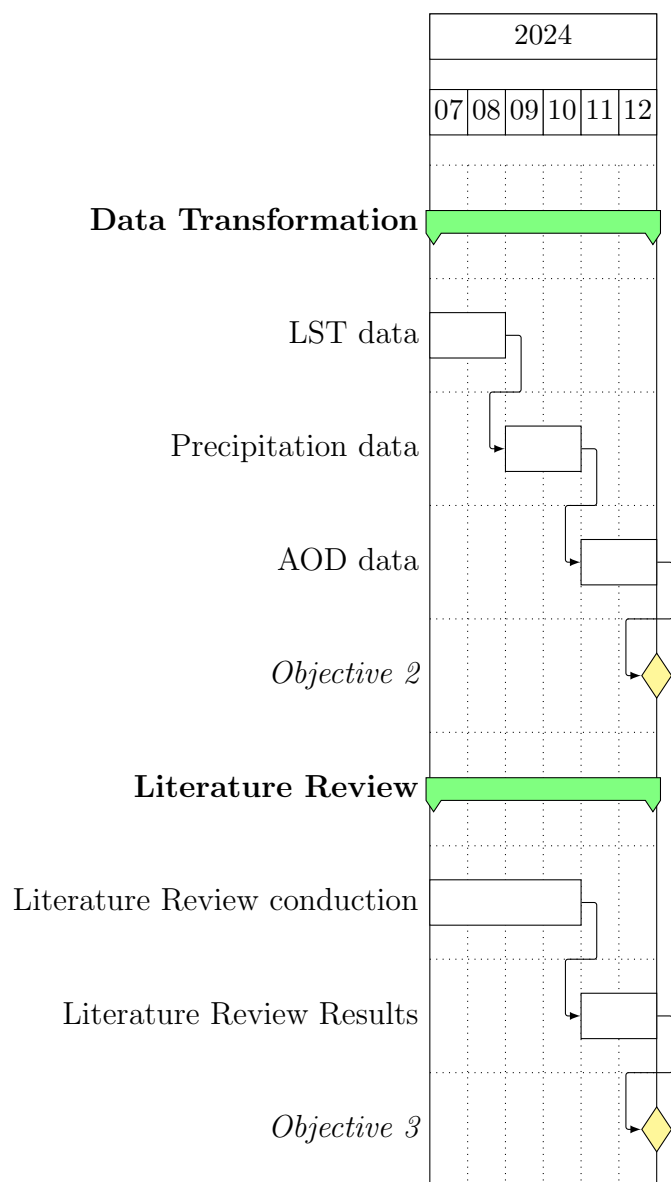


Figure 6: Activities to follow until December, 2024.

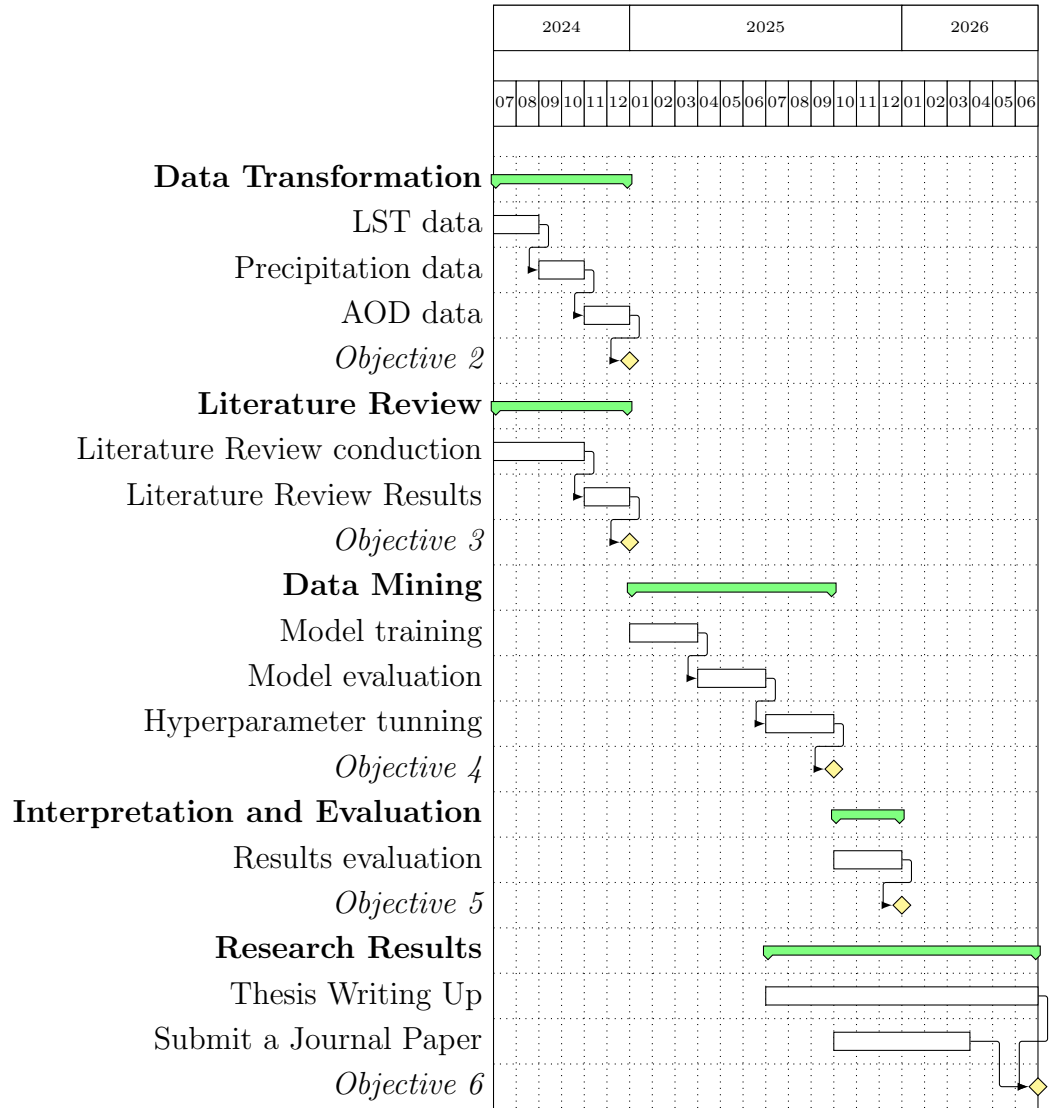


Figure 7: Activities to follow to achieve the Research Goal (From July, 2024 To June, 2026).

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